

NIST Trustworthy and Responsible AI

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Adversarial Machine Learning

A Taxonomy and Terminology of Attacks and Mitigations

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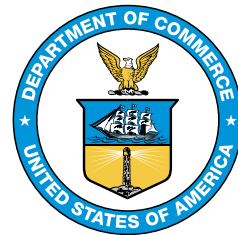
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Abstract

This NIST Trustworthy and Responsible AI report provides a taxonomy of concepts and defines terminology in the field of adversarial machine learning (AML). The taxonomy is arranged in a conceptual hierarchy that includes key types of ML methods, life cycle stages of attack, and attacker goals, objectives, capabilities, and knowledge. This report also identifies current challenges in the life cycle of AI systems and describes corresponding methods for mitigating and managing the consequences of those attacks. The terminology used in this report is consistent with the literature on AML and is complemented by a glossary of key terms associated with the security of AI systems. Taken together, the taxonomy and terminology are meant to inform other standards and future practice guides for assessing and managing the security of AI systems by establishing a common language for the rapidly developing AML landscape.

Keywords

artificial intelligence; machine learning; attack taxonomy; abuse; data poisoning; evasion; privacy breach; attack mitigation; large language model; chatbot.

NIST Trustworthy and Responsible AI

The National Institute of Standards and Technology (NIST) promotes U.S. innovation and industrial competitiveness by advancing measurement science, standards, and technology in ways that enhance economic security and improve our quality of life. Among its broad range of activities, NIST contributes to the research, standards, evaluations, and data required to advance the development, use, and assurance of trustworthy artificial intelligence (AI).

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Audience

The intended primary audience for this document includes individuals and groups who are responsible for designing, developing, deploying, evaluating, and governing AI systems.

Background

This document is the result of an extensive literature review, conversations with experts in adversarial machine learning, and research performed by the authors in adversarial machine learning.

Trademark Information

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The Information Technology Laboratory (ITL) at NIST develops tests, test methods, reference data, proof of concept implementations, and technical analyses to advance the development and productive use of information technology. ITL's responsibilities include the development of management, administrative, technical, and physical standards and guidelines.

This NIST Trustworthy and Responsible AI report focuses on identifying, addressing, and managing risks associated with adversarial machine learning. While practical guidance¹ published by NIST may serve as an informative reference, this guidance remains voluntary.

The content of this document reflects recommended practices. This document is not intended to serve as or supersede existing regulations, laws, or other mandatory guidance.

¹In the context of this paper, the terms “practice guide,” “guide,” “guidance,” and the like are consensus-created informative references that are intended for voluntary use. They should not be interpreted as equal to the use of the term “guidance” in a legal or regulatory context. This document does not establish any legal standard or any other legal requirement or defense under any law, nor does it have the force or effect of law.

How to Read This Document

This document uses the terms “AI technology,” “AI system,” and “AI applications” interchangeably. Terms related to the machine learning pipeline, such as “ML model” or “algorithm,” are also used interchangeably in this document. Depending on context, the term “system” may refer to the broader organizational and/or social ecosystem within which the technology was designed, developed, deployed, and used instead of the more traditional use related to computational hardware or software.

Important reading notes:

- This document includes a series of blue callout boxes that highlight nuances and important takeaways.
- This document contains links shown in blue. Clicking on them will bring the reader to the relevant resource. Links in the References point to external sources.
- Terms that are used but not defined or explained in the text are listed and defined in the Glossary. They are displayed in small caps in the text. Clicking on a word shown in SMALL CAPS (e.g., [ADVERSARIAL EXAMPLE](#)) takes the reader directly to the definition of that term in the Glossary. From there, one may click on the page number shown at the end of the definition to return.
- This document provides an [Index](#) of attack types to easily navigate and reference attacks and corresponding mitigations.

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Author Contributions

The authors contributed equally to this work.

Predictive AI and Generative AI Taxonomy Index

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 - * Model Poisoning (ID: [NISTAML.051](#))

Executive Summary

This NIST Trustworthy and Responsible AI report describes a taxonomy and terminology for **ADVERSARIAL MACHINE LEARNING** (AML) that may aid in securing applications of artificial intelligence (AI) against adversarial manipulations and attacks.

The statistical, data-based nature of ML systems opens up new potential vectors for attacks against these systems' security, privacy, and safety, beyond the threats faced by traditional software systems. These challenges span different phases of ML operations such as the potential for adversarial manipulation of training data; the provision of adversarial inputs to adversely affect the performance of the AI system; and even malicious manipulations, modifications, or interactions with models to exfiltrate sensitive information from the model's training data or to which the model has access. Such attacks have been demonstrated under real-world conditions, and their sophistication and impacts have been increasing steadily.

The field of AML is concerned with studying these attacks. It must consider the capabilities of attackers, the model or system properties that attackers might seek to violate in pursuit of their objectives, and the design of attack methods that exploit vulnerabilities during the development, training, and deployment phases of the ML life cycle. It is also concerned with the design of ML algorithms and systems that can withstand these security and privacy challenges, a property often known as robustness [274].

To taxonomize these attacks, this report differentiates between predictive and generative AI systems and the attacks relevant to each. It considers the components of an AI system including the data; the model itself; the processes for training, testing, and deploying the model; and the broader software and system contexts into which models may be embedded, such as cases where Generative Artificial Intelligence (GenAI) models are deployed with access to private data or equipped with tools to take actions with real-world consequences.

Thus, the attacks within this taxonomy are classified relative to: (i) the AI system type, (ii) the stage of the ML life cycle process in which the attack is mounted, (iii) the attacker's goals and objectives in terms of the system properties they seek to violate, (iv) the attacker's capabilities and access, and (v) the attacker's knowledge of the learning process and beyond.

This report adopts the concepts of security, resilience, and robustness of ML systems from the NIST AI Risk Management Framework. Security, resilience, and robustness are gauged by risk, which is a measure of the extent to which an entity (e.g., a system) is threatened by a potential circumstance or event (e.g., an attack) and the severity of the outcome should such an event occur. However, this report does not make recommendations on risk tolerance (i.e., the level of risk that is acceptable to organizations or society) because it is highly contextual and specific to applications and use cases.

The spectrum of effective attacks against ML is wide, rapidly evolving, and covers all phases of the ML lifecycle — from design and implementation to training, testing, and deployment in the real world. The nature and power of these attacks are different and their impacts may depend not only on the vulnerabilities of the ML models but also the weaknesses of the infrastructure in which the AI systems are deployed. AI system components may also be adversely affected by design and implementation flaws that cause failures outside the context of adversarial use, such as inaccuracy. However, these kinds of flaws are not within the scope of the literature on AML or the attacks in this report.

In addition to defining a taxonomy of attacks, this report provides corresponding methods for mitigating and managing the consequences of those attacks in the life cycle of AI systems, and outlines the limitations of widely used mitigation techniques to raise awareness and help organizations increase the efficacy of their AI risk-mitigation efforts. The terminology used in this report is consistent with the literature on AML and is complemented by a glossary that defines key terms associated with the field of AML in order to assist non-expert readers. Taken together, the taxonomy and terminology are meant to inform other standards and future practice guides for assessing and managing the security of AI systems by establishing a common language for the rapidly developing AML landscape. Like the taxonomy, the terminology and definitions are not intended to be exhaustive but rather to serve as a starting point for understanding and aligning on key concepts that have emerged in the AML literature.

1. Introduction

Artificial intelligence (AI) systems have been on a global expansion trajectory for several years [267]. These systems are being developed by and widely deployed into the economies of numerous countries, with increasing opportunities for people to use AI systems in many spheres of their lives [92]. This report distinguishes between two broad classes of AI systems: predictive AI (PredAI) and generative AI (GenAI). Although the majority of industrial applications of AI systems are still dominated by PredAI systems, there has been a recent increase in the adoption of GenAI systems in business and consumer contexts. As these systems permeate the digital economy and become essential parts of daily life, the need for their secure, robust, and resilient operation grows. These operational attributes are critical elements of trustworthy AI in the NIST AI Risk Management Framework [274] and the NCSC Machine Learning Principles [266].

The field of **ADVERSARIAL MACHINE LEARNING** (AML) studies attacks against ML systems that exploit the statistical, data-based nature of ML systems. Despite the significant progress of AI and machine learning (ML) in different application domains, these technologies remain vulnerable to attacks that can cause spectacular failures. The chances of these kinds of failure increase as ML systems are used in contexts where they may be subject to novel or adversarial interactions, and the consequences grow more dire as these systems are used in increasingly high-stakes domains. For example, in PredAI computer vision applications for object detection and classification, well-known cases of adversarial perturbations of input images have caused autonomous vehicles to swerve into lanes going in the opposite direction, stop signs to be misclassified as speed limit signs, and even people wearing glasses to be misidentified in high-security settings [121, 187, 332, 349]. Similarly, the potential for adversarial input to trick ML models into revealing hidden information has become more urgent as more ML models are being deployed in fields like medicine, where medical record leaks can expose sensitive personal information [25, 171].

In GenAI, large language models (LLMs) [13, 15, 49, 85, 102, 236, 247, 277, 279, 348, 365, 371, 372, 436] are increasingly becoming an integral part of software applications and internet infrastructure. LLMs are being used to create more powerful online search tools, help software developers write code, and power chatbots that are used by millions of people every day [255]. LLMs are also being augmented to create more useful AI systems, including through interactions with corporate databases and documents to enable powerful **RETRIEVAL-AUGMENTED GENERATION (RAG)** (RAG) [210] and through training- or inference-time techniques to enable LLMs to take real-world actions, such as browsing the web or using a bash terminal as an LLM-based **AGENT** [167, 261, 278, 419]. Thus, vulnerabilities in GenAI systems may expose a broad attack surface for threats to the privacy of sensitive user data or proprietary information about models' architecture or training data, and create risks to the integrity and availability of widely used systems.

As GenAI adoption has grown, the increasing capability of these systems has created another challenge for model developers: how to manage the risks created by unwanted or

harmful uses of these systems' capabilities.[275] As model developers have increasingly sought to apply technical interventions to reduce models' potential for misuse, another surface for high-stakes AML attacks has emerged in attacks that attempt to circumvent or disrupt these protections.

Fundamentally, many AI systems are susceptible both to AML attacks and to attacks that more closely resemble traditional cybersecurity attacks, including attacks against the platforms on which they are deployed. This report focuses on the former and considers the latter to be within the scope of traditional cybersecurity taxonomies.

Both PredAI and GenAI systems are vulnerable to attacks enabled by a range of attacker capabilities throughout the development and deployment life cycle. Attackers can manipulate training data [327], including the Internet data used in large-scale model training [57], or can modify test-time inference data and resources by adding adversarial perturbations or suffixes. Attackers can also attack the components used to make AI systems by inserting **TROJAN** functionality. As organizations increasingly rely on pre-trained models that could be used directly or fine-tuned with new datasets to enable different tasks, their vulnerability to these attacks increases.

Modern cryptography often relies on algorithms that are secure in an information-theoretic sense, that is, those that can be formally proven to ensure security under certain conditions. However, there are no information-theoretic security proofs for the widely used ML algorithms in modern AI systems. Moreover, information-theoretic *impossibility* results that set limits on the effectiveness of widely used mitigation techniques have begun to appear in the literature [124, 140, 432]. As a result, many of the advances in developing mitigations against different classes of AML attacks tend to be empirical and limited in nature, adopted because they appear to work in practice rather than because they provide information-theoretic security guarantees. Thus, many of these mitigations may themselves be vulnerable to new discoveries and evolutions in attacker techniques.

This report offers guidance for the development of:

- Standardized terminology for AML terms that can be used across relevant ML and cybersecurity communities. There are notable differences in terminology in different stakeholder communities and it is important to work towards bridging the differences as AI is increasingly adopted throughout enterprise and consumer contexts.
- A taxonomy of the most widely studied and currently effective attacks in AML, including:
 - Evasion, poisoning, and privacy attacks for PredAI systems
 - Poisoning, direct prompting, and indirect prompt injection attacks for GenAI systems
- A discussion of potential mitigations for these attacks and the limitations of existing mitigation techniques

NIST intends to update this report as new developments emerge in AML attacks and mitigations.

This report provides a categorization of common classes of attacks and their mitigations for PredAI and GenAI systems. This report is not intended to provide an exhaustive survey of all available literature on Adversarial ML, which includes more than 11,354 references on [arXiv.org](https://arxiv.org) since 2021 as of July 2024.

This report is organized into three sections.

- Section [2](#) considers PredAI systems. Section [2.1](#) introduces the taxonomy of attacks for PredAI systems, which defines the broad categories of attacker objectives and goals, and identifies the capabilities that an adversary must leverage to achieve the corresponding objectives. Specific attack classes are also introduced for each type of capability. Sections [2.2](#), [2.3](#), and [2.4](#) discuss the major classes of attacks: evasion, poisoning, and privacy, respectively. A corresponding set of mitigations for each class of attacks is provided in the attack class sections.
- Section [3](#) considers GenAI systems. Section [3.1](#) introduces the taxonomy of attacks for GenAI systems and defines the broad categories of attacker objectives and adversary capabilities relevant to these systems. Specific attack classes are introduced for each type of capability, along with relevant mitigations.
- Section [4](#) discusses remaining challenges in the field, including limitations to widely used mitigation techniques. The intent is to raise awareness of open questions in the field of AML and to call attention to trends that may shape risk and risk management practices in future.

2. Predictive AI Taxonomy

2.1. Attack Classification

Figure 1 introduces a taxonomy of attacks in AML on PredAI systems, based on attacker goals and objectives, capabilities, and knowledge.

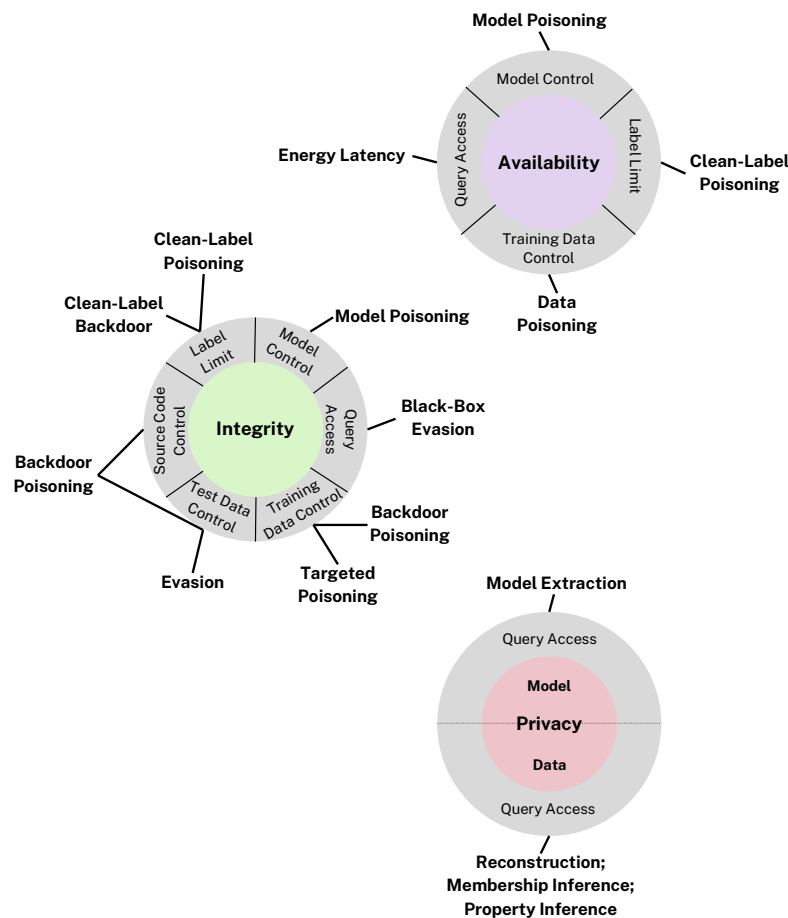


Figure 1. Taxonomy of attacks on PredAI systems

The attacker's objectives are shown as disjointed circles with the attacker's goal at the center of each circle: **availability** breakdown, **integrity** violation, and **privacy** compromise. The capabilities that an adversary must leverage to achieve their objectives are shown in the outer layer of the objective circles. Attack classes are shown as callouts connected to the capabilities required to mount each attack. Multiple attack classes that require the same capabilities to reach the same objective are shown in a single callout.

These attacks are classified according to the following dimensions: 1) learning method and stage of the learning process when the attack is mounted, 2) attacker goals and objectives,

3) attacker capabilities, and 4) attacker knowledge of the learning process. Several adversarial attack classification frameworks have been introduced in prior works [42, 358], and the goal here is to create a standard terminology for adversarial attacks on ML that unifies existing work.

2.1.1. Stages of Learning

Predictive machine learning involves a **TRAINING STAGE** in which a model is learned and a **DEPLOYMENT STAGE** in which the model is deployed on new, unlabeled data samples to generate predictions. In the case of **SUPERVISED LEARNING**, labeled training data is given as input to a training algorithm in the training stage, and the ML model is optimized to minimize a specific loss function. Validation and testing of the ML model is usually performed before the model is deployed in the real world. Common supervised learning techniques include **CLASSIFICATION** in which the predicted labels or *classes* are discrete and **REGRESSION** in which the predicted labels or *response variables* are continuous.

Other learning paradigms in the ML literature include **UNSUPERVISED LEARNING**, which trains models using unlabeled data at training time; **SEMI-SUPERVISED LEARNING** in which a small set of examples have labels, while the majority of samples are unlabeled; **REINFORCEMENT LEARNING** in which an agent interacts with an environment and learns an optimal policy to maximize its reward; **FEDERATED LEARNING** in which a set of clients jointly train an ML model by communicating with a server that performs an aggregation of model updates; and **ENSEMBLE LEARNING**, which is an approach that seeks better predictive performance by combining the predictions from multiple models.

Most PredAI models are **DISCRIMINATIVE**, i.e., learn only a decision boundary, such as **LOGISTIC REGRESSION**, **SUPPORT VECTOR MACHINES**, and **CONVOLUTIONAL NEURAL NETWORKS**. GenAI models may also be used in predictive tasks, such as sentiment analysis [125].

AML literature predominantly considers adversarial attacks against AI systems that could occur at either the training stage or the deployment stage. During the training stage, the attacker might control part of the training data, their labels, the model parameters, or the code of ML algorithms, resulting in different types of poisoning attacks. During the deployment stage, the ML model is already trained, and the adversary could mount evasion attacks to create integrity violations and change the ML model's predictions, as well as privacy attacks to infer sensitive information about the training data or the ML model.

Training-time attacks. **POISONING ATTACKS** [40] occur during the ML training stage. In a **DATA POISONING** attack [40, 148], an adversary controls a subset of the training data by either inserting or modifying training samples. In a **MODEL POISONING** attack [222], the adversary controls the model and its parameters. Data poisoning attacks are applicable to all learning paradigms, while model poisoning attacks are most prevalent in federated learning [190], where clients send local model updates to the aggregating server, and in supply-chain attacks, where malicious code may be added to the model by suppliers of

model technology.

Deployment-time attacks. Other types of attacks can be mounted against deployed models. Evasion attacks modify testing samples to create **ADVERSARIAL EXAMPLE** [38, 144, 362], which are similar to the original sample (e.g., according to certain distance metrics) but alter the model predictions to the attacker's choices. Other attacks, such as availability attacks and privacy attacks including membership inference [342] and data reconstruction [110], can also be mounted by attackers with query access to a deployed ML model.

2.1.2. Attacker Goals and Objectives

The attacker's objectives are classified along three dimensions according to the three main types of security violations considered when analyzing the security of a system: availability breakdown, integrity violation, and privacy compromise. Figure 1 separates attacks into three disjointed circles according to their objective, and the attacker's objective is shown at the center of each circle.

Availability breakdown [NISTAML.01] [Back to Index]. An **AVAILABILITY BREAKDOWN** attack is a deliberate interference with a PredAI system to disrupt the ability of other users or processes to obtain timely and reliable access to its services. This attack type may be initiated at training or deployment time, although its impacts are typically experienced at deployment time. Availability attacks can be mounted via data poisoning, when the attacker controls a fraction of the training set; via model poisoning, when the attacker controls the model parameters; or as **ENERGY-LATENCY ATTACK** via query access. Data poisoning availability attacks have been proposed for **SUPPORT VECTOR MACHINES** [40], linear regression [179], and even neural networks [228, 260], while model poisoning attacks have been designed for neural networks [222] and federated learning [22].

- **Energy latency attacks [NISTAML.014] [Back to Index].** Recently, **ENERGY-LATENCY ATTACK**, a type of availability attacks that require only black-box access to the model, have been developed for neural networks across many different tasks in computer vision and natural language processing (NLP) [345].

Integrity violation [NISTAML.02] [Back to Index]. An **INTEGRITY VIOLATION** attack is a deliberate interference with a PredAI system to force it to misperform against its intended objectives and produce predictions that align with the adversary's objective. An attacker can cause an integrity violation by mounting an evasion attack at deployment time or a poisoning attack at training time. Evasion attacks require the modification of testing samples to create adversarial examples that are misclassified by the model while often remaining stealthy and imperceptible to humans [38, 144, 362]. Integrity attacks via poisoning can be classified as **TARGETED POISONING ATTACK** [137, 330], **BACKDOOR POISONING ATTACK** [148], and **MODEL POISONING** [22, 36, 123]. Targeted poisoning tries to violate the integrity of a few targeted samples and assumes that the attacker has training data control to insert the poisoned samples. Backdoor poisoning attacks require the generation of a **BACKDOOR PATTERN**,

which is added to both the poisoned samples and the testing samples to cause misclassification. Backdoor attacks are the only attacks in the literature that require both training and testing data control. Model poisoning attacks could result in either targeted or backdoor attacks, and the attacker modifies model parameters to cause an integrity violation. They have been designed for centralized learning [222] and federated learning [22, 36].

Privacy compromise [NISTAML.03] [Back to Index]. A **PRIVACY COMPROMISE** attack causes the unintended leakage of restricted or proprietary information from a PredAI system, including details about a model’s training data, weights, or architecture [100, 309]. While the term “confidentiality” is more widely used in taxonomies of traditional cybersecurity attacks, the AML field has tended to use the top-level term “privacy” to encompass both attacks against the confidentiality of a model (e.g., those that extract information about a model’s weights or architecture) and those that cause violations of expected privacy properties of model outputs (e.g. by exposing model training data) [310]. **DATA CONFIDENTIALITY** during ML training can be achieved through secure computation methods based on cryptographic techniques [2, 253, 288, 385], which ensure that training data and model parameters remain protected during the training phase. However, even models trained using paradigms that enforce data confidentiality may be vulnerable to privacy attacks, in which adversaries interacting with a model can extract information about its training data or parameters. In this report, we focus on privacy compromises that can occur at deployment time, regardless of the training method used, or whether data confidentiality was maintained during training.

In privacy attacks, attackers might be interested in learning information about the training data (resulting in **DATA PRIVACY ATTACKS**) or the ML model (resulting in **MODEL PRIVACY ATTACKS**). The attacker could have different objectives for compromising the privacy of training data, such as **DATA RECONSTRUCTION** [110] (inferring the content or features of training data), **MEMBERSHIP-INFERENC ATTACK** [162, 343] (inferring the presence of data in the training set), **TRAINING DATA EXTRACTION** [59, 63] (extracting training data from generative models), **ATTRIBUTE INFERENC ATTACKS** [184, 409] (inferring sensitive attributes of training records) and **PROPERTY INFERENC** [134] (inferring properties about the training data distribution). **MODEL EXTRACTION** is a model privacy attack in which attackers aim to extract information about the model [177].

2.1.3. Attacker Capabilities

AML attacks for PredAI systems can be taxonomized with respect to the capabilities that an attacker controls. An adversary might leverage six types of capabilities to achieve their objectives, as shown in the outer layer of the objective circles in Fig. 1:

- **TRAINING DATA CONTROL**: The attacker might take control of a subset of the training data by inserting or modifying training samples. This capability is used in data poisoning attacks (e.g., availability poisoning, targeted or backdoor poisoning).

- **MODEL CONTROL:** The attacker might take control of the model parameters by either generating a Trojan trigger and inserting it in the model or by sending malicious local model updates in federated learning.
- **TESTING DATA CONTROL:** The attacker might add perturbations to testing samples at model deployment time, as performed in evasion attacks to generate adversarial examples or in backdoor poisoning attacks.
- **LABEL LIMIT:** This capability is relevant to restrict adversarial control over the labels of training samples in supervised learning. Clean-label poisoning attacks assume that the attacker does not control the label of the poisoned samples, while regular poisoning attacks assume label control over the poisoned samples.
- **SOURCE CODE CONTROL:** The attacker might modify the source code of the ML algorithm, such as the random number generator or any third-party libraries, which are often open source.
- **QUERY ACCESS:** The attacker might submit queries to the model and receive predictions (i.e., labels or model confidences), such as when interacting with an AI system hosted by a cloud provider as a machine learning as a service (MLaaS) offering. This capability is used by black-box evasion attacks, **ENERGY-LATENCY ATTACK**, and all privacy attacks that do not require knowledge of the model's training data, architecture, or parameters.

Even if an attacker does not have the ability to modify training/testing data, source code, or model parameters, access to these may still be crucial for mounting stronger white-box attacks that require knowledge of the ML system. See Sec. 2.1.4 for more details on attacker knowledge, and detailed definitions of white-box and black-box attacks.

Figure 1 connects each attack class with the capabilities required to mount the attack. For example, backdoor attacks that cause integrity violations require control of the training and testing data to insert the backdoor pattern. Backdoor attacks can also be mounted via source code control, particularly when training is outsourced to a more powerful entity. Clean-label backdoor attacks do not allow label control on the poisoned samples in addition to the capabilities needed for backdoor attacks.

2.1.4. Attacker Knowledge

Another dimension of attack classification is how much knowledge the attacker has about the ML system. There are three main types of attacks:

White-box attacks. These assume that the attacker operates with *full* knowledge about the ML system, including the training data, model architecture, and model hyperparameters. While these attacks operate under very strong assumptions, the main reason for analyzing them is to test the vulnerability of a system against worst-case adversaries and

to evaluate potential mitigations. This definition is more general and encompasses the notion of adaptive attacks in which knowledge of the mitigations applied to the model or the system is explicitly tracked.

Black-box attacks. These attacks assume that the attacker operates with minimal, and sometimes no knowledge at all about the ML system. An adversary might have query access to the model, but they have no other information about how the model is trained. These attacks are the most practical since they assume that the attacker has no knowledge of the AI system and utilizes system interfaces readily available for normal use.

Gray-box attacks. There are a range of gray-box attacks that capture adversarial knowledge between black-box and white-box attacks. Suciu et al. [358] introduced a framework to classify gray-box attacks. An attacker might know the model architecture but not its parameters, or the attacker might know the model and its parameters but not the training data. Other common assumptions for gray-box attacks are that the attacker has access to data distributed identically to the training data and knows the feature representation. The latter assumption is important for applications in which feature extraction is used before training an ML model, such as cybersecurity, finance, and healthcare.

2.1.5. Data Modality

Until recently, most attacks and defenses in adversarial machine learning have operated under a single modality, but a new trend in the field is to use multimodal data. The taxonomy of attacks defined in Fig. 1 is independent of the modality of the data in specific applications.

The most common data modalities in the AML literature include:

- **Image:** Adversarial examples of image data [144, 362] have the advantage of a continuous domain, and gradient-based methods can be applied directly for optimization. Backdoor poisoning attacks were first invented for images [148], and many privacy attacks are run on image datasets (e.g., [342]). The image modality includes other types of imaging (e.g., LIDAR, SAR, IR, hyperspectral).
- **Text:** Text is a popular modality, and all classes of attacks have been proposed for text models, including evasion [150], poisoning [82, 213], and privacy [426].
- **Audio:** Audio systems and text generated from audio signals have also been attacked [66].
- **Video:** Video comprehension models have shown increasing capabilities in vision and language tasks [428], but such models are also vulnerable to attacks [402].
- **Cybersecurity**²: The first poisoning attacks were discovered in cybersecurity for worm

²Cybersecurity data may not include a single modality but rather multiple modalities, such as network-level, host-level, or program-level data.

signature generation (2006) [291] and spam email classification (2008) [269]. Since then, poisoning attacks have been shown for malware classification, malicious PDF detection, and Android malicious app classification [329]. Evasion attacks against similar data modalities have been proposed as well: malware classification [103, 357], PDF malware classification [352, 414], Android malicious app detection [295], and network intrusion detection [93]. Poisoning unsupervised learning models has been shown for clustering used in malware classification [41] and network traffic anomaly detection [315].

Anomaly detection based on data-centric approaches allows for automated feature learning through ML algorithms. However, the application of ML to such problems comes with specific challenges related to the need for very low false negative and low false positive rates (e.g., the ability to catch zero-day attacks). This challenge is compounded by the fact that trying to accommodate all of these together makes ML models susceptible to adversarial attacks [198, 301, 446].

- **Tabular data:** There have been numerous attacks against ML models working on tabular data, such as poisoning availability attacks against healthcare and business applications [179], privacy attacks against healthcare data [422], and evasion attacks against financial applications [141].

Recently, the use of ML models trained on multimodal data has gained traction, particularly the combination of image and text data modalities. Several papers have shown that multimodal models may provide some resilience against attacks [417], but other papers show that multimodal models themselves could be vulnerable to attacks mounted on all modalities at the same time [77, 333, 415] (see Sec. 4.2.3).

An open challenge is to test and characterize the resilience of a variety of multimodal ML models against evasion, poisoning, and privacy attacks.

2.2. Evasion Attacks and Mitigations

[NISTAML.022] [Back to Index]

The discovery of evasion attacks against ML models has led to significant growth in AML research over the last decade. In an evasion attack, the adversary’s goal is to generate adversarial examples: samples whose classification can be changed to an arbitrary class of the attacker’s choice – often with only minimal perturbation [362]. For example, in the context of image classification, the perturbation of the original sample might be small so that a human cannot observe the transformation of the input; while the ML model can be tricked to classify the adversarial example in the target class selected by the attacker, humans still recognize it as part of the original class.

Early known instances of evasion attacks date back to 1988 with the work of Kearns and Li [192] and 2004 when Dalvi et al. [98] and Lowd and Meek [226] demonstrated the existence of adversarial examples for linear classifiers used in spam filters. Later, Szegedy et al. [362] showed that deep neural networks used for image classification could be easily manipulated through adversarial examples. In 2013, Szegedy et al. [362] and Biggio et al. [38] independently discovered an effective method for generating adversarial examples against linear models and neural networks by applying gradient optimization to an adversarial objective function. Both of these techniques require white-box access to the model and were improved by subsequent methods that generated adversarial examples with even smaller perturbations [20, 65, 232].

Adversarial examples are also applicable in more realistic black-box settings in which attackers only obtain query access capabilities to the trained model. Even in the more challenging black-box setting in which attackers obtain the model’s predicted labels or confidence scores, deep neural networks are still vulnerable to adversarial examples. Methods for creating adversarial examples in black-box settings include zeroth-order optimization [80], discrete optimization [254], Bayesian optimization [344], and *transferability*, which involves the white-box generation of adversarial examples on a different model before transferring them to the target model [282, 283, 377]. While cybersecurity and image classifications were the first application domains to showcase evasion attacks, ML technology in many other application domains has come under scrutiny, including speech recognition [66], natural language processing [185], and video classification [215, 401].

Mitigating adversarial examples is a well-known challenge in the community and deserves additional research and investigation. The field has a history of publishing defenses evaluated under relatively weak adversarial models that are subsequently broken by more powerful attacks. Mitigations need to be evaluated against strong adaptive attacks, and guidelines for the rigorous evaluation of newly proposed mitigation techniques have been established [97, 375]. The most promising directions for mitigating the critical threat of evasion attacks are adversarial training [144, 232] (iteratively generating and inserting adversarial examples with their correct labels at training time); certified techniques, such as

randomized smoothing [94] (evaluating ML prediction under noise); and formal verification techniques [136, 191] (applying formal method techniques to verify the model’s output). Nevertheless, these methods have different limitations, such as decreased accuracy for adversarial training and randomized smoothing and computational complexity for formal methods. There is an inherent trade-off between robustness and accuracy [374, 379, 433]. Similarly, there are trade-offs between a model’s robustness and fairness guarantees [71].

2.2.1. White-Box Evasion Attacks

In the white-box threat model, the attacker has full knowledge of the model architecture and parameters, as discussed in Section 2.1.4. The main challenge for creating adversarial examples in this setting is to find a perturbation added to a testing sample that changes its classification label, often with constraints on properties such as the perceptibility or size of the perturbation. In the white-box threat model, it is common to craft adversarial examples by solving an optimization problem written from the attacker’s perspective, which specifies the objective function for the optimization (such as changing the target label to a certain class), as well as a distance metric to measure the similarity between the testing sample and the adversarial example.

Optimization-based methods. Szedegy et al. [362] and Biggio et al. [38] independently proposed the use of optimization techniques to generate adversarial examples. In their threat models, the adversary is allowed to inspect the entirety of the ML model and compute gradients relative to the model’s loss function. These attacks can be targeted (i.e., the adversarial example’s class is selected by the attacker) or untargeted (i.e., the adversarial examples are misclassified to any other incorrect class).

Szedegy et al. [362] coined the widely used term *adversarial examples*. They considered an objective that minimized the ℓ_2 norm of the perturbation subject to the model prediction changing to the target class. The optimization is solved using the limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) method. Biggio et al. [38] considered the setting of a binary classifier with malicious and benign classes with a continuous and differentiable discriminant function. The objective of the optimization is to minimize the discriminant function in order to generate adversarial examples of maximum confidence.

While Biggio et al. [38] applied their method to linear classifiers, kernel SVM, and multi-layer perceptrons, Szedegy et al. [362] showed the existence of adversarial examples on deep learning models used for image classification. Goodfellow et al. [144] introduced an efficient method for generating adversarial examples for deep learning: the Fast Gradient Sign Method (FGSM), which performs a single iteration of gradient descent for solving the optimization. This method has been extended to an iterative FGSM attack by Kurakin et al. [200].

Subsequent works have proposed new objectives and methods for optimizing the generation of adversarial examples with the goals of minimizing the perturbations and supporting

multiple distance metrics. Some notable attacks include:

- DeepFool is an untargeted evasion attack for ℓ_2 norms, which uses a linear approximation of the neural network to construct the adversarial examples [257].
- The Carlini-Wagner attack uses multiple objectives that minimize the loss or logits on the target class and the distance between the adversarial example and original sample. The attack is optimized via the penalty method [65] and considers three distance metrics to measure the perturbations of adversarial examples: ℓ_0 , ℓ_2 , and ℓ_∞ . The attack has been effective against the defensive distillation defense [284].
- The Projected Gradient Descent (PGD) attack [232] minimizes the loss function and projects the adversarial examples to the space of allowed perturbations at each iteration of gradient descent. PGD can be applied to the ℓ_2 and ℓ_∞ distance metrics for measuring the perturbation of adversarial examples.

Universal evasion attacks. Moosavi-Dezfooli et al. [256] showed how to construct small universal perturbations (with respect to some norm) that can be added to most images and induce a misclassification. Their technique relies on successive optimization of the universal perturbation using a set of points sampled from the data distribution. This is a form of FUNCTIONAL ATTACK. An interesting observation is that the universal perturbations generalize across deep network architectures, suggesting similarity in the decision boundaries trained by different models for the same task.

Physically realizable attacks. These are attacks against ML systems that can be implemented feasibly in the physical world [21, 200, 227]. One of the first instances was the attack on facial recognition systems by Sharif et al. [332]. The attack can be realized by printing a pair of eyeglass frames, which misleads facial recognition systems to either evade detection or impersonate another individual. Eykholt et al. [122] proposed an attack to generate robust perturbations under different conditions, resulting in adversarial examples that can evade vision classifiers in various physical environments. The attack is applied to evade a road sign detection classifier by physically applying black and white stickers to the road signs. The ShapeShifter [81] attack was designed to evade object detectors, which is a more challenging problem than attacking image classifiers since the attacker needs to evade the classification in multiple bounding boxes with different scales. This attack also requires the perturbation to be robust enough to survive real-world distortions due to different viewing distances, angles, lighting conditions, and camera limitations.

Other data modalities. In computer vision applications, adversarial examples are often designed to be imperceptible to humans. Therefore, the perturbations introduced by attackers need to be so small that a human correctly recognizes the images, while the ML classifier is tricked into changing its prediction. Alternatively, there may be a trigger object in the image that is still imperceptible or innocuous to humans but causes the model to misclassify. The concept of adversarial examples has been extended to other domains, such as audio, video, NLP, and cybersecurity. In some of these settings, there are additional

constraints that need to be respected by adversarial examples, such as text semantics in NLP and the application constraints in cybersecurity. Several representative works include:

- **Audio:** Carlini and Wagner [66] showed a targeted attack on models that generate text from speech. They can generate an audio waveform that is very similar to an existing one but that can be transcribed to any text of the attacker's choice.
- **Video:** Adversarial evasion attacks against video classification models can be split into sparse attacks that perturb a small number of video frames [401] and dense attacks that perturb all of the frames in a video [215]. The goal of the attacker is to change the classification label of the video.
- **Text:** Jia and Liang [185] developed a methodology for generating adversarial text examples. This pioneering work was followed by many advances in developing adversarial attacks on natural language processing (NLP) models (see a comprehensive survey on the topic [438]). La Malfa and Kwiatkowska [202] proposed a method for formalizing perturbation definitions in NLP by introducing the concept of semantic robustness. The main challenges in NLP are that the domain is discrete rather than continuous (e.g., image, audio, and video classification), and adversarial examples need to respect text semantics. These challenges are illustrated by the recent ASCII-art attack [186] against chatbots. An ASCII-art illustration of a forbidden term tricks the chatbot into providing the harmful information even when the chatbot correctly censors the plain English word. The semantic distance between the two prompts is precisely zero, and both of them should have been treated the same.
- **Cybersecurity:** In cybersecurity applications, adversarial examples must respect the constraints imposed by the application semantics and feature representation of cyber data, such as network traffic or program binaries. FENCE is a general framework for crafting white-box evasion attacks using gradient optimization in discrete domains and supports a range of linear and statistical feature dependencies [88]. FENCE has been applied to two network security applications: malicious domain detection and malicious network traffic classification. Sheatsley et al. [334] proposed a method that learns the constraints in feature space using formal logic and crafts adversarial examples by projecting them onto a constraint-compliant space. They applied the technique to network intrusion detection and phishing classifiers. Both papers observed that attacks from continuous domains cannot be readily applied in constrained environments, as they result in infeasible adversarial examples. Pierazzi et al. [295] discussed the difficulty of mounting feasible evasion attacks in cybersecurity due to constraints in feature space and the challenge of mapping attacks from feature space to problem space. They formalized evasion attacks in problem space and constructed feasible adversarial examples for Android malware.

2.2.2. Black-Box Evasion Attacks

[NISTAML.025] [Back to Index]

Black-box evasion attacks are designed under a realistic adversarial model in which the attacker has no prior knowledge of the model architecture or training data. Instead, the adversary can interact with a trained ML model by querying it on various data samples and obtaining the model's predictions. Similar APIs are provided by MLaaS offered by public cloud providers, in which users can obtain the model's predictions on selected queries without information about how the model was trained. There are two main classes of black-box evasion attacks in the literature:

- **Score-based attacks:** In this setting, attackers obtain the model's confidence scores or logits and can use various optimization techniques to create the adversarial examples. A popular method is zeroth-order optimization, which estimates the model's gradients without explicitly computing derivatives [80, 173]. Other optimization techniques include discrete optimization [254], natural evolution strategies [172], and random walks [262].
- **Decision-based attacks:** In this more restrictive setting, attackers only obtain the final predicted labels of the model. The first method for generating evasion attacks was the Boundary Attack based on random walks along the decision boundary and rejection sampling [47], which was extended with an improved gradient estimation to reduce the number of queries in the HopSkipJumpAttack [79]. More recently, several optimization methods search for the direction of the nearest decision boundary (e.g., the OPT attack [86]), use sign SGD instead of binary searches (e.g., the Sign-OPT attack [87]), or use Bayesian optimization [344].

The primary challenge in creating adversarial examples in black-box settings is reducing the number of queries to the ML models. Recent techniques can successfully evade the ML classifiers with a relatively small number of queries, typically less than 1000 [344].

2.2.3. Transferability of Attacks

Another method for generating adversarial attacks under restrictive threat models involves transferring an attack crafted on a different ML model. Typically, an attacker trains a substitute ML model, generates white-box adversarial attacks on the substitute model, and transfers the attacks to the target model. Various methods differ in how the substitute models are trained. For example, Papernot et al. [282, 283] train the substitute model with score-based queries to the target model, while several papers train an ensemble of models without explicitly querying the target model [218, 377, 397].

Attack transferability is an intriguing phenomenon, and existing literature attempts to un-

derstand the fundamental reasons why adversarial examples transfer across models. Several papers have observed that different models learn intersecting decision boundaries in both benign and adversarial dimensions, which leads to better transferability [144, 256, 377]. Demontis et al. [104] identified two main factors that contribute to attack transferability for both evasion and poisoning: the intrinsic adversarial vulnerability of the target model and the complexity of the surrogate model used to optimize the attack. EXPECTATION OVER TRANSFORMATION aims to make adversarial examples sustain image transformations that occur in the real world, such as angle and viewpoint changes [21].

2.2.4. Evasion attacks in the real world

While many of the attacks discussed in this section were demonstrated only in research settings, several evasion attacks have been demonstrated in the real world, and we discuss prominent instances in face recognition systems, phishing webpage detection, and malware classification.

Face recognition systems used for identity verification have been the target of adversarial evasion attacks, as they constitute an entry point to critical systems and enable users to commit financial fraud. During the last half of 2020, the ID.me face recognition service found more than 80,000 attempts of users attempting to fool their ID verification steps used by multiple state workforce agencies [276]. These attacks included people wearing masks, using deepfakes, or using images or videos of other people. The intent was to fraudulently claim unemployment benefits provided during COVID relief efforts. Later in 2022, according to US federal prosecutors, a New Jersey man was able to verify fake driver's licenses through ID.me as part of a US\$ 2.5M unemployment-fraud scheme. This time, the suspect used various wigs to evade the face recognition system [156].

Another case study of real-world evasion attacks reported by Apruzzese et al. [17] is an attack against a commercial phishing webpage detector. The ML phishing detector is an ensemble of multiple models that analyze different aspects of the image to determine if it is a phishing attempt. Inputs that are marked uncertain by the model are triaged to security analysts. Out of 4600 samples marked uncertain by the ML image classification system, the authors identified 100 adversarial examples. Interestingly, a manual analysis of these adversarial examples revealed that attackers do not employ optimization-based attacks, but rather utilize relatively simple methods for evasion, such as image cropping, masking, or blurring techniques.

Other examples of evasion attacks demonstrated by researchers in malware classification are cataloged in the MITRE Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS) knowledge base [248]. Palo Alto Networks reported evasion attacks against a deep learning detector for malware command-and-control traffic, and a botnet Domain Generation Algorithm (DGA) detector. An instance of a universal evasion attack was discovered against Cylance's AI malware detection model. Researchers also evaded ProofPoint's email protection system by training a shadow ML model and using the insights from that to at-

tack the real system. These are demonstrations of evasion vulnerabilities by researchers, but did not result in attacks in the wild.

2.2.5. Mitigations

Mitigating evasion attacks is challenging because adversarial examples are widespread in a variety of ML model architectures and application domains. Possible explanations for the existence of adversarial examples are that ML models rely on non-robust features that are not aligned with human perception in the computer vision domain [174]. In the last few years, many of the proposed mitigations against adversarial examples have been ineffective against stronger attacks. Furthermore, several papers have performed extensive evaluations and defeated a large number of proposed mitigations:

- Carlini and Wagner showed how to bypass 10 methods for detecting adversarial examples and described several guidelines for evaluating defenses [64]. Recent work shows that detecting adversarial examples is as difficult as building a defense [373]. Therefore, this direction for mitigating adversarial examples is similarly challenging as designing defenses.
- The Obfuscated Gradients attack [20] was specifically designed to defeat several proposed defenses that rely on masking gradients to protect against optimization-based attacks. It relies on a new technique, Backward Pass Differentiable Approximation, which approximates the gradient during the backward pass of backpropagation, and was shown to bypass several proposed defenses based on gradient masking.
- Tramèr et al. [375] described a methodology for designing adaptive attacks against proposed defenses and circumvented 13 existing defenses. They advocate for designing adaptive attacks to test newly proposed defenses rather than merely testing the defenses against well-known attacks.

From the wide range of proposed defenses against adversarial evasion attacks, three main classes have proven to be resilient and have the potential to provide mitigation against evasion attacks:

1. **Adversarial training:** Introduced by Goodfellow et al. [144] and further developed by Madry et al. [232], adversarial training is a general method that augments training data with adversarial examples generated iteratively during training using their correct labels. The stronger the adversarial attacks for generating adversarial examples are, the more resilient the trained model becomes. Adversarial training results in models with more semantic meaning than standard models [379], but this benefit usually comes at the cost of decreased model accuracy on clean data. Additionally, adversarial training is expensive due to the iterative generation of adversarial examples during training.
2. **Randomized smoothing:** Proposed by Lecuyer et al. [207] and further improved by

Cohen et al. [94], randomized smoothing is a method that transforms any classifier into a certifiable robust smooth classifier by producing the most likely predictions under Gaussian noise perturbations. This method results in provable robustness for ℓ_2 evasion attacks, even for classifiers trained on large-scale datasets, such as ImageNet. Randomized smoothing typically provides certified prediction to a subset of testing samples, the exact number of which depends on factors such as the size of the potential perturbations or the characteristics of the training data and model. Recent results have extended the notion of certified adversarial robustness to ℓ_2 -norm bounded perturbations by combining a pretrained denoising diffusion probabilistic model and a standard high-accuracy classifier [62]. Li et al. [211] developed a taxonomy for the robustness verification and training of representative algorithms. They also revealed the characteristics, strengths, limitations, and fundamental connections among these approaches, along with theoretical barriers facing the field.

3. **Formal verification:** Another method for certifying the adversarial robustness of a neural network is based on techniques from FORMAL METHODS. Reluplex uses satisfiability modulo theories (SMT) solvers to verify the robustness of small feed-forward neural networks [191]. AI² is the first verification method applicable to convolutional neural networks using abstract interpretation techniques [136]. These methods have been extended and scaled up to larger networks in follow-up verification systems, such as DeepPoly [346], ReluVal [394], and Fast Geometric Projections (FGP) [131]. Formal verification techniques have significant potential for certifying neural network robustness but are limited by their lack of scalability, computational cost, and restriction in the type of supported algebraic operations such as addition, multiplication, etc.

All of these proposed mitigations exhibit inherent trade-offs between robustness and accuracy, and they come with additional computational costs during training. Therefore, designing ML models that resist evasion while maintaining accuracy remains an open problem. See Section 4.1.1 for further discussion on these trade-offs.

2.3. Poisoning Attacks and Mitigations

Poisoning attacks are broadly defined as adversarial attacks during the training stage of the ML algorithm. The first known poisoning attack was developed for worm signature generation in 2006 [291]. Since then, poisoning attacks have been studied extensively in several application domains: computer security (for spam detection [269], network intrusion detection [384], vulnerability prediction [318], malware classification [329, 412]), computer vision [137, 148, 330], natural language processing (NLP) [82, 213, 388], and tabular data in healthcare and financial domains [179]. Recently, poisoning attacks have gained more attention in industry applications as well [199]. They can even be orchestrated at scale so that an adversary with limited financial resources could control a fraction of the public datasets used for model training [57].

Poisoning attacks are powerful and can cause availability or integrity violations. Availability poisoning attacks typically cause indiscriminate degradation of the ML model on all samples, while targeted and backdoor poisoning attacks induce integrity violations on a small set of target samples. Poisoning attacks leverage a wide range of adversarial capabilities (e.g., data poisoning, model poisoning, label control, source code control, and test data control), resulting in several subcategories of poisoning attacks. They have been developed in white-box [40, 179, 412], gray-box [179], and black-box settings [39].

This section describes availability poisoning, targeted poisoning, backdoor poisoning, and model poisoning attacks classified according to their adversarial objective. For each poisoning attack category, techniques for mounting the attacks, existing mitigations, and their limitations are also discussed. The classification of poisoning attacks in this document is inspired by the framework developed by Cinà et al. [91], which includes additional references to poisoning attacks and mitigations.

2.3.1. Availability Poisoning

[NISTAML.013] [Back to Index]

The first poisoning attacks discovered in cybersecurity applications were availability attacks against worm signature generation and spam classifiers, which indiscriminately degrade the performance of the entire ML model in order to effectively prevent its use. Perdisci et al. [291] generated suspicious flows with fake invariants that mislead the worm signature generation algorithm in Polygraph [270]. Nelson et al. [269] designed poisoning attacks against Bayes-based spam classifiers by generating training samples of “spam” emails containing long sequences of words that appear in legitimate emails, degrading the performance of the spam classifier by inducing a higher rate of false positives. Both of these attacks were conducted under the white-box setting in which adversaries were aware of the ML training algorithm, feature representations, training datasets, and ML models. Availability poisoning attacks have also been proposed for ML-based systems that detect cybersecurity attacks against industrial control systems: such detectors are often retrained

using data collected during system operation to account for plant operational drift of the monitored signals, creating opportunities for an attacker to mimic the signals of corrupted sensors at training time to poison the detector such that real attacks remain undetected at deployment time [198].

A simple black-box poisoning attack strategy is LABEL FLIPPING, in which an adversary generates training examples with incorrect or altered labels [39]. This method may require a large percentage of poisoning samples to mount an availability attack. These attacks can also be formulated through optimization-based methods, such as by solving a bilevel optimization problem to determine the optimal poisoning samples that will achieve the adversarial objective (i.e., maximize the hinge loss for a SVM [40] or maximize the mean square error [MSE] for regression [179]). Similar optimization-based availability poisoning attacks have been designed against linear regression [179] and neural networks [260], although these optimization-based attacks may require white-box access to the model and training data. In gray-box adversarial settings, the most popular method for generating availability poisoning attacks is transferability, in which poisoning samples are generated for a surrogate model and transferred to the target model [104, 358].

Clean-label poisoning [NISTAML.012] [Back to Index]. A realistic threat model for supervised learning is that of clean-label poisoning attacks, in which adversaries can only control the training examples but not their labels. This may arise in scenarios in which the labeling process is external to the training algorithm, as in malware classification where binary files can be submitted by attackers to threat intelligence platforms and labeling is performed using anti-virus signatures or other external methods. Clean-label availability attacks have been introduced for neural network classifiers by training a generative model and adding noise to training samples to maximize the adversarial objective [128]. A different approach for clean-label poisoning is to use gradient alignment and minimally modify the training data [129].

Availability poisoning attacks have also been designed for unsupervised learning against centroid-based anomaly detection [195] and behavioral clustering for malware [41]. In federated learning, an adversary can mount a model poisoning attack to induce availability violations in the globally trained model [123, 335, 336]. More details on model poisoning attacks are provided in Sec. 2.3.

Mitigations. Availability poisoning attacks are usually detectable by monitoring the standard performance metrics of ML models (e.g., precision, recall, accuracy, F1 scores, and area under the curve) as they cause a large degradation in the classifier metrics. However, detecting these attacks during the testing or deployment stages of ML may be less desirable, and many existing mitigations aim to proactively prevent these attacks during the training stage to generate robust ML models. Existing mitigations for availability poisoning attacks include:

- **Training data sanitization:** These methods leverage the insight that poisoned samples are typically different than regular training samples that are not controlled by

adversaries. As such, data sanitization techniques are designed to clean the training set and remove the poisoned samples before the ML training is performed. Cretu et al. [96] proposed the first sanitization procedure for unlabeled datasets that relies on majority voting of multiple models trained on subsets of the training set. They apply the method to anomaly detection on network packets. Nelson et al. [269] introduced the Region of Non-Interest (RONI) method, which examines each sample and excludes it from training if the accuracy of the model decreases when the sample is added. Subsequently proposed sanitization methods improved upon these early approaches by reducing their computational complexity and considering other applications. Paudice et al. [289] introduced a method for label cleaning that was specifically designed for label-flipping attacks. Steinhardt et al. [354] proposed the use of outlier detection methods for identifying poisoned samples. Clustering methods have also been used to detect poisoned samples [203, 363]. Other work has suggested that computing the variance of predictions made by an ensemble of multiple ML models is an effective data sanitization method for network intrusion detection [384]. Once sanitized, datasets may be protected by cybersecurity mechanisms for provenance and integrity attestation [267].

- **Robust training:** An alternative approach to mitigating availability poisoning attacks is to modify the ML training algorithm to increase the robustness of the resulting model. The defender can train an ensemble of multiple models and generate predictions via model voting [37, 209, 395]. Several papers apply techniques from robust optimization, such as using a trimmed loss function [109, 179]. Rosenfeld et al. [314] proposed the use of randomized smoothing to add noise during training to provide protection against label-flipping attacks.

2.3.2. Targeted Poisoning

[NISTAML.024] [Back to Index]

In contrast to availability attacks, targeted poisoning attacks induce a change in the ML model's prediction on a small number of targeted samples. If the adversary can control the labeling function of the training data, then label-flipping is an effective targeted poisoning attack: the adversary simply inserts several poisoned samples with the target label, and the model will learn the wrong label. Therefore, targeted poisoning attacks are mostly studied in a clean-label setting in which the attacker does not have control over training data labels.

Several techniques for mounting clean-label targeted attacks have been proposed. Koh and Liang [196] showed how influence functions (i.e., a statistical method that determines the most influential training samples for a prediction) can be leveraged to create poisoned samples in the fine-tuning setting in which a pre-trained model is fine-tuned on new data. Suciu et al. [358] designed StingRay, a targeted poisoning attack that modifies samples in feature space and adds poisoned samples to each mini batch of training. An optimization proce-

ture based on feature collision was crafted by Shafahi et al. [330] to generate clean-label targeted poisoning for fine-tuning and end-to-end learning. ConvexPolytope [444] and BullseyePolytope [4] optimized the poisoning samples against ensemble models, which offers better advantages for attack transferability. MetaPoison [166] uses a meta-learning algorithm to optimize the poisoned samples, while Witches' Brew [137] performs optimization by gradient alignment, resulting in a state-of-the-art targeted poisoning attack.

All of the above attacks impact a small set of targeted samples that are selected by the attacker during training, and they have only been tested for continuous image datasets (with the exception of StingRay, which requires adversarial control of a large fraction of the training set). Subpopulation poisoning attacks [180] were designed to poison samples from an entire subpopulation, defined by matching on a subset of features or creating clusters in representation space. Poisoned samples are generated using label-flipping (for NLP and tabular modalities) or a first-order optimization method (for continuous data, such as images). The attack generalizes to all samples in a subpopulation and requires minimal knowledge about the ML model and a small number of poisoned samples proportional to the subpopulation size.

Targeted poisoning attacks have also been introduced for semi-supervised learning algorithms [53], such as MixMatch [34], FixMatch [347], and Unsupervised Data Augmentation (UDA) [413] in which the adversary poisons a small fraction of the unlabeled training dataset to change the prediction on targeted samples at deployment time.

Mitigations. Targeted poisoning attacks are notoriously challenging to defend against. Jagielski et al. [180] showed an impossibility result for subpopulation poisoning attacks. To mitigate some of the risks associated with such attacks, model developers may protect training data through traditional cybersecurity measures such as access controls, use methods for data sanitization and validation, and use mechanisms for dataset provenance and integrity attestation [267]. Ma et al. [230] proposed the use of differential privacy (DP) as a defense (which follows directly from the definition of differential privacy), but differentially private ML models may also have lower accuracy than standard models, and the trade-off between robustness and accuracy needs to be considered in each application. See Section 4.1.1 for further discussion on the trade-offs between the attributes of Trustworthy AI systems.

2.3.3. Backdoor Poisoning

[NISTAML.021, NISTAML.023] [Back to Index]

Backdoor poisoning attacks are poisoning attacks that cause the targeted model to misclassify samples containing a particular BACKDOOR PATTERN or trigger. In 2017, Gu et al. [148] proposed BadNets, the first backdoor poisoning attack. They observed that image classifiers can be poisoned by adding a small patch trigger in a subset of images at training time and changing their label to a target class. The classifier learns to associate the trigger

with the target class, and any image that includes the trigger or backdoor pattern will be misclassified to the target class at testing time. Concurrently, Chen et al. [84] introduced backdoor attacks in which the trigger is blended into the training data. Follow-up work introduced the concept of clean-label backdoor attacks [380] in which the adversary cannot change the label of the poisoned examples. Clean-label attacks typically require more poisoning samples to be effective, but the attack model is more realistic.

In the last few years, backdoor attacks have become more sophisticated and stealthy, making them harder to detect and mitigate. Latent backdoor attacks were designed to survive even upon model fine-tuning of the last few layers using clean data [420]. Backdoor Generating Network (BaN) [322] is a dynamic backdoor attack in which the location of the trigger changes in the poisoned samples so that the model learns the trigger in a location-invariant manner. Functional triggers (i.e., FUNCTIONAL ATTACK) are embedded throughout the image or change according to the input. Li et al. used steganography algorithms to hide the trigger in the training data [214] and introduced a clean-label attack that uses natural reflection on images as a backdoor trigger [223]. Wenger et al. [404] poisoned facial recognition systems by using physical objects as triggers, such as sunglasses and earrings. Architectural backdoor attacks [205] perform malicious modifications to the structure of an ML model during its training phase, which allows an attacker to manipulate the model's behavior when presented with specific triggers. These attacks require adversarial access to the model design or training environment and are applicable when model training is outsourced to a more powerful entity, such as a cloud service.

Other data modalities. While the majority of backdoor poisoning attacks are designed for computer vision applications, this attack vector has been effective in other application domains with different data modalities, such as audio, NLP, and cybersecurity settings.

- **Audio:** In audio domains, Shi et al. [341] showed how an adversary can inject an unnoticeable audio trigger into live speech, which is jointly optimized with the target model during training.
- **NLP:** In NLP, the construction of meaningful poisoning samples is more challenging as the text data is discrete, and the semantic meaning of sentences would ideally be preserved for the attack to remain unnoticeable. Recent work has shown that backdoor attacks in NLP domains are becoming feasible. For instance, Chen et al. [82] introduced semantic-preserving backdoors at the character, word, and sentence level for sentiment analysis and neural machine translation applications. Li et al. [213] generated hidden backdoors against transformer models using generative language models in three NLP tasks: toxic comment detection, neural machine translation, and question answering.
- **Cybersecurity:** Following early work on poisoning in cybersecurity [269, 291], Severi et al. [329] showed how AI explainability techniques can be leveraged to generate clean-label poisoning attacks with small triggers against malware classifiers. They attacked multiple models (i.e., neural networks, gradient boosting, random forests,

and SVMs) using three malware datasets: Ember for Windows PE file classification, Contagio for PDF file classification, and DREBIN for Android app classification. Jigsaw Puzzle [418] designed a backdoor poisoning attack for Android malware classifiers that uses realizable software triggers harvested from benign code.

Mitigations. The literature on backdoor attack mitigation is vast compared to other poisoning attacks. Below we discuss several classes of defenses, including data sanitization, trigger reconstruction, and model inspection and sanitization, and we also mention their limitations.

- **Training data sanitization:** Similar to poisoning availability attacks, training data sanitization can be applied to detecting backdoor poisoning attacks. For example, outlier detection in the latent feature space [157, 293, 378] has been effective for convolutional neural networks used for computer vision applications. Activation Clustering [76] clusters training data in representation space to isolate the backdoored samples in a separate cluster. Data sanitization achieves better results when the poisoning attack controls a relatively large fraction of training data but is not as effective against stealthy poisoning attacks. Overall, this leads to a trade-off between attack success and the detectability of malicious samples.
- **Trigger reconstruction:** This class of mitigations aims to reconstruct the backdoor trigger, assuming that it is at a fixed location in the poisoned training samples. NeuralCleanse by Wang et al. [390] developed the first trigger reconstruction approach and used optimization to determine the most likely backdoor pattern that reliably misclassifies the test samples. The initial technique has been improved to reduce performance time on several classes and simultaneously support multiple triggers inserted into the model [163, 411]. A representative system in this class is Artificial Brain Simulation (ABS) by Liu et al. [221], which stimulates multiple neurons and measures the activations to reconstruct the trigger patterns. Khaddaj et al. [193] developed a new primitive for detecting backdoor attacks and a corresponding effective detection algorithm with theoretical guarantees.
- **Model inspection and sanitization:** Model inspection analyzes the trained ML model before its deployment to determine whether it was poisoned. An early work in this space is NeuronInspect [168], which is based on explainability methods to determine different features between clean and backdoored models that are subsequently used for outlier detection. DeepInspect [78] uses a conditional generative model to learn the probability distribution of trigger patterns and performs model patching to remove the trigger. Xu et al. [416] proposed the Meta Neural Trojan Detection (MNTD) framework, which trains a meta-classifier to predict whether a given ML model is backdoored (or “Trojaned,” in the authors’ terminology). This technique is general and can be applied to multiple data modalities, such as vision, speech, tabular data, and NLP. Once a backdoor is detected, model sanitization can be performed via pruning [407], retraining [429], or fine-tuning [217] to restore the

model's accuracy.

- **Certified defenses:** Several methods for achieving certified defenses against data poisoning attacks have been proposed in the literature. BagFlip [440] is a model-agnostic defense that extends randomized smoothing [94] and combines training data bagging with adding noise to both training and testing samples. Deep Partition Aggregation [209] and Deep Finite Aggregation [396] are certified defenses that partition the training data into disjointed subsets and train an ensemble method on each partition to reduce the impact of poisoned samples. Recently, FCert [398] provides a certified defense against data poisoning in few-shot classification settings used for both vision and text data.

Most of these mitigations have been designed against computer vision classifiers based on convolutional neural networks using backdoors with fixed trigger patterns. Severi et al. [329] showed that some of the data sanitization techniques (e.g., spectral signatures [378] and Activation Clustering [76]) are ineffective against clean-label backdoor poisoning on malware classifiers. More recent semantic and functional backdoor triggers would also pose challenges to approaches based on trigger reconstruction or model inspection, which generally assume fixed backdoor patterns. The limitation of using meta classifiers for predicting a Trojanged model [416] is the high computational complexity of the training stage of the meta classifier, which requires training thousands of SHADOW MODEL. Additional research is required to design strong backdoor mitigation strategies that can protect ML models against this important attack vector without suffering from these limitations.

In cybersecurity, Rubinstein et al. [315] proposed an approach based on principal component analysis (PCA) to mitigate poisoning attacks against PCA subspace anomaly detection methods in backbone networks. It maximized median absolute deviation (MAD) instead of variance to compute principal components and used a threshold value based on Laplace distribution instead of Gaussian. Madani and Vlajic [231] built an autoencoder-based intrusion detection system, assuming that malicious poisoning attack instances were under 2%.

[193] provided a different perspective on backdoor mitigation by showing that backdoors are indistinguishable from naturally occurring features in the data if no additional assumptions are made about the attack. However, assuming that the backdoor creates the strongest feature in the data, the paper proposed an optimization technique to identify and remove the training samples that correspond to the backdoor.

Poison forensics [331] is a technique for root cause analysis that identifies malicious training samples and complements existing mitigations that are not always resilient in the face of evolving attacks. Poison forensics adds another layer of defense in an ML system: once a poisoning attack is detected at deployment time, poison forensics can trace back to the source of the attack in the training set.

2.3.4. Model Poisoning

[NISTAML.011, NISTAML.026] [Back to Index]

Model poisoning attacks attempt to directly modify the trained ML model to inject malicious functionality into it. In centralized learning, TrojNN [222] reverse engineers the trigger from a trained neural network and then retrains the model by embedding the trigger in external data to poison it. Most model poisoning attacks have been designed in the federated learning setting in which clients send local model updates to a server that aggregates them into a global model. Compromised clients can send malicious updates to poison the global model. Model poisoning attacks can cause both availability and integrity violation in federated models:

- Poisoning availability attacks that degrade the global model’s accuracy have been effective, but they usually require a large percentage of clients to be under the control of the adversary [123, 335].
- Targeted model poisoning attacks induce integrity violations on a small set of samples at testing time. They can be mounted by a model replacement or model boosting attack in which the compromised client replaces the local model update according to the targeted objective [23, 35, 360].
- Backdoor model poisoning attacks introduce a trigger via malicious client updates to induce the misclassification of all samples with the trigger at testing time [23, 35, 360, 392]. Most of these backdoors are forgotten if the compromised clients do not regularly participate in training, but the backdoor becomes more durable if injected in the lowest utilized model parameters [441].

Supply chain model poisoning. [NISTAML.05] [NISTAML.051] [Back to Index] Model poisoning attacks are also possible in supply-chain scenarios in which models or components of the model provided by suppliers are poisoned with malicious code. Dropout Attack [425] is a recent supply-chain attack that shows how an adversary who manipulates the randomness used in neural network training (particularly in dropout regularization) might poison the model to decrease accuracy, precision, or recall on a set of targeted classes. See Supply Chain Attacks and Mitigations for additional discussion of supply-chain risks to GenAI models that are applicable to PredAI models too.

Mitigations. A variety of Byzantine-resilient aggregation rules have been designed and evaluated to defend federated learning from model poisoning attacks. Most of them attempt to identify and exclude the malicious updates when performing the aggregation at the server [8, 43, 51, 149, 242–244, 359, 423]. However, motivated adversaries can bypass these defenses by adding constraints to the attack generation optimization problem [23, 123, 335]. Gradient clipping and differential privacy have the potential to mitigate model poisoning attacks to some extent [23, 271, 360], but they usually decrease accuracy and do not provide complete mitigation.

For specific model poisoning vulnerabilities, such as backdoor attacks, there are some techniques for model inspection and sanitization (see Sec. 2.3.3). However, mitigating supply-chain attacks in which adversaries might control the source code of the training algorithm or the ML hyperparameters remains challenging. Program verification techniques used in other domains (e.g., cryptographic protocol verification [299]) might be adapted to this setting, but ML algorithms have intrinsic randomness and non-deterministic behavior, which enhances the difficulty of verification.

Designing ML models that are robust in the face of supply-chain model poisoning vulnerabilities is a critical open problem.

2.3.5. Poisoning Attacks in the Real World

As poisoning attacks require adversarial control over the ML training process, they are difficult to mount in the real world. Still, there are several examples of documented cases of real poisoning attacks targeting early AI chatbots, email spam filters, and malware classification services.

The first example of a real-world poisoning attack is the Tay.AI chatbot, a chatbot released by Microsoft on Twitter in 2016 [272]. After online interaction with users for less than 24 hours, the chatbot was poisoned and immediately taken down. At about the same time, there were several large-scale efforts to compromise Google's Gmail spam filter, in which attackers sent millions of emails to attempt to poison the Gmail spam classifier algorithm, enabling them to send other malicious emails without being detected [272]. MITRE ATLAS reported a poisoning incident on the VirusTotal threat intelligence service, in which similar, but not identical samples of a ransomware family were submitted through a popular virus sharing platform to cause the mis-classification of that particular ransomware family [248].

These incidents highlight the risks associated with online learning, as the Tay.AI chatbot was updated in real-time based on user interactions, and the Gmail spam filter and the VirusTotal malware classification system were continuously updated based on newly received samples. In all these incidents, attackers crafted poisoned samples after an initial model release, counting on the fact that models are continuously updated.

2.4. Privacy Attacks and Mitigations

[NISTAML.03] [Back to Index]

The seminal work of Dinur and Nissim [110] introduced DATA RECONSTRUCTION attacks, which seek to reverse-engineer private information about an individual user record or other sensitive input data from access to a trained model. More recently, data reconstruction attacks have been designed for binary and multi-class neural network classifiers [50, 152]. With MEMBERSHIP-INFERENC ATTACK, an adversary can determine whether a particular record was included in the dataset used for training an ML model. Membership inference attacks were first introduced by Homer et al. [162] for genomic data. Recent literature focuses on membership attacks against ML models in mostly black-box settings in which adversaries have query access to a trained ML model [54, 342, 422]. Property inference attacks [19, 74, 134, 233, 361, 437] aim to extract global information about a training dataset, such as the fraction of training examples with a certain sensitive attribute. A different privacy violation for MLaaS is MODEL EXTRACTION attacks, which are designed to extract information about an ML model, such as its architecture or model parameters³ [58, 70, 177, 376].

This section discusses privacy attacks related to data reconstruction, the memorization of training data, membership inference, property inference, and model extraction, as well as mitigations for some of these attacks and open problems in designing general mitigation strategies.

2.4.1. Data Reconstruction

[NISTAML.032] [Back to Index]

Data reconstruction attacks have the ability to recover an individual's data from released aggregate information. Dinur and Nissim [110] were the first to introduce reconstruction attacks that recover user data from linear statistics. Their original attack required an exponential number of queries for reconstruction, but subsequent work has shown how to perform reconstruction with a polynomial number of queries [116]. A survey of privacy attacks, including reconstruction attacks, is given by Dwork et al. [114]. More recently, the U.S. Census Bureau performed a large-scale study on the risk of data reconstruction attacks on census data [135], which motivated the use of differential privacy in the decennial release of the U.S. Census in 2020.

In the context of ML classifiers, Fredrickson et al. [130] introduced model inversion attacks that reconstruct class representatives from the training data of an ML model. While

³A privacy violation in this context describes a loss of confidential information about an ML model. If ML model leakage leads to further privacy violations for individuals (e.g., identity theft, sensitive data extraction), it may also be viewed as a cybersecurity-related privacy event. For further discussion on the relationship between privacy and cybersecurity risks, see NIST Privacy Framework, Version 1.0.

model inversion generates semantically similar images as those in the training set, it cannot directly reconstruct the training data of the model. Recently, Balle et al. [26] trained a reconstructor network that can recover a data sample from a neural network model, assuming that a powerful adversary has information about all other training samples. Haim et al. [152] showed how the training data of a binary neural network classifier can be reconstructed from access to the model parameters by leveraging theoretical insights about implicit bias in neural networks. This work has recently been extended to reconstruct training samples of multi-class multi-layer perceptron classifiers [50]. Attribute inference is another relevant privacy attack in which the attacker extracts a sensitive attribute of the training set, assuming partial knowledge about other features in the training data [184].

The ability to reconstruct training samples is partially explained by the tendency of neural networks to memorize their training data. Zhang et al. [431] discussed how neural networks can memorize randomly selected datasets. Feldman [126] showed that the memorization of training labels is necessary to achieve an almost optimal generalization error in ML. Brown et al. [48] constructed two learning tasks based on next-symbol prediction and cluster labeling in which memorization is required for high-accuracy learning. Feldman and Zhang empirically evaluated the benefit of memorization for generalization using an influence estimation method [127]. Data reconstruction attacks and their connection to memorization for generative AI are discussed in Sec. 3.3.2.

2.4.2. Membership Inference

[NISTAML.033] [Back to Index]

Membership inference attacks may expose private information about an individual, like reconstruction or memorization attacks, and are of great concern when releasing aggregate information or ML models trained on user data. In certain situations, determining that an individual is part of the training set already has privacy implications, such as in a medical study of patients with a rare disease. Moreover, membership inference can be used as a building block for mounting data extraction attacks [59, 63].

In membership inference, the attacker’s goal is to determine whether a particular record or data sample was part of the training dataset used for the statistical or ML algorithm. These attacks were introduced by Homer et al. [162] for statistical computations on genomic data under the name *tracing attacks*. Robust tracing attacks have been analyzed when an adversary gains access to noisy statistical information about the dataset [115]. In the last five years, the literature has used the terminology *membership inference* for attacks against ML models. Most of the attacks in the literature are performed against deep neural networks that are used for classification [54, 89, 208, 342, 421, 422]. Similar to other attacks in AML, membership inference can be performed in white-box settings [208, 264, 317] in which attackers have knowledge of the model’s architecture and parameters, but most of the attacks have been developed for black-box settings in which the adversary generates queries to the trained ML model [54, 89, 342, 421, 422].

The attacker's success in membership inference has been formally defined using a cryptographically inspired privacy game in which the attacker interacts with a challenger and needs to determine whether a target sample was used in training the queried ML model [183, 321, 422]. In terms of techniques for mounting membership inference attacks, the loss-based attack by Yeom et al. [422] is one of the most efficient and widely used method. Using the knowledge that the ML model minimizes the loss on training samples, the attack determines that a target sample is part of training if its loss is lower than a fixed threshold (selected as the average loss of training examples). Sablayrolles et al. [317] refined the loss-based attack by scaling the loss using a per-example threshold. Another popular technique introduced by Shokri et al. [342] is *shadow models*, which trains a meta-classifier on examples in and out of the training set obtained by training thousands of shadow ML models on the same task as the original model. This technique is generally expensive, and while it might improve upon the simple loss-based attack, its computational cost is high and requires access to many samples from the distribution to train the shadow models. These two techniques are at opposite ends of the spectrum in terms of their complexity, but they perform similarly in terms of precision at low false positive rates [54].

An intermediary method that obtains good performance in terms of the AREA UNDER THE CURVE (AUC) metric is the LiRA attack by Carlini et al. [54], which trains a smaller number of shadow models to learn the distribution of model logits on examples in and out of the training set. Using the assumption that the model logit distributions are Gaussian, LiRA performs a hypothesis test for membership inference by estimating the mean and standard deviation of the Gaussian distributions. Ye et al. [421] designed a similar attack that performs a one-sided hypothesis test, which does not make any assumptions on the loss distribution but achieves slightly lower performance than LiRA. Recently, Lopez et al. [225] proposed a more efficient membership inference attack that requires training a single model to predict the quantiles of the confidence score distribution of the model under attack. Membership inference attacks have also been designed under the stricter label-only threat model in which the adversary only has access to the predicted labels of the queried samples [89].

There are several public privacy libraries that offer implementations of membership inference attacks: the TensorFlow Privacy library [350] and the ML Privacy Meter [259].

2.4.3. Property Inference

[NISTAML.034] [Back to Index]

In property inference attacks (also called distribution inference), the attacker tries to learn global information about the training data distribution by interacting with an ML model. For example, an attacker can determine the fraction of the training set with a certain sensitive attribute (e.g., demographic information) that might reveal potentially confidential information about the training set that is not intended to be released.

Property inference attacks were introduced by Ateniese et al. [19] and formalized as a distinguishing game between the attacker and the challenger training two models with different fractions of the sensitive data [361]. Property inference attacks were designed in white-box settings in which the attacker has access to the full ML model [19, 134, 361] and black-box settings in which the attacker issues queries to the model and learns either the predicted labels [233] or the class probabilities [74, 437]. These attacks have been demonstrated for HIDDEN MARKOV MODEL, SUPPORT VECTOR MACHINES [19], FEEDFORWARD NEURAL NETWORKS [134, 233, 437], CONVOLUTIONAL NEURAL NETWORKS [361], FEDERATED LEARNING [240], GENERATIVE ADVERSARIAL NETWORKS [443], and GRAPH NEURAL NETWORK [442]. Mahloujifar et al. [233] and Chaudhuri et al. [74] showed that poisoning the property of interest can help design a more effective distinguishing test for property inference. Moreover, Chaudhuri et al. [74] designed an efficient property size estimation attack that recovers the exact fraction of the population of interest.

The relationship between different training set inference attacks, such as membership inference, attribute inference, and property inference, has been explored by Salem et al. [321] under a unified definitional framework.

2.4.4. Model Extraction

[NISTAML.031] [Back to Index]

In MLaaS scenarios, cloud providers typically train large ML models using proprietary data and would like to keep the model architecture and parameters confidential. The goal of an attacker performing a MODEL EXTRACTION attack is to extract information about the model architecture and parameters by submitting queries to the ML model trained by an MLaaS provider. The first model stealing attacks were shown by Tramer et al. [376] on several online ML services for different ML models, including logistic regression, decision trees, and neural networks. However, Jagielski et al. [177] have shown the exact extraction of ML models to be impossible. Instead, a functionally equivalent model can be reconstructed that is different than the original model but achieves similar performance at the prediction task. Jagielski et al. [177] have shown that even the weaker task of extracting functionally equivalent models is computationally prohibitive (*NP-hard*).

Several techniques for mounting model extraction attacks have been introduced in the literature. The first method is that of direct extraction based on the mathematical formulation of the operations performed in deep neural networks, which allows the adversary to compute model weights algebraically [58, 177, 376]. A second technique is to use learning methods for extraction. For example, active learning [70] can guide the queries to the ML model for more efficient extraction of model weights, and reinforcement learning can train an adaptive strategy that reduces the number of queries [280]. A third technique uses SIDE CHANNEL information for model extraction. Batina et al. [29] used electromagnetic side channels to recover simple neural network models, while Rakin et al. [303] showed how ROWHAMMER ATTACK can be used for model extraction of more complex convolutional

neural network architectures.

Model extraction is often not an end goal but a step toward other attacks. As the model weights and architecture become known, attackers can launch more powerful attacks that are typical for the white-box or gray-box settings. Therefore, preventing model extraction can mitigate downstream attacks that depend on the attacker having knowledge of the model architecture and weights.

2.4.5. Mitigations

The discovery of reconstruction attacks against aggregate information motivated the rigorous definition of *differential privacy* (DP) [112, 113], an extremely strong definition of privacy that guarantees a bound on how much an attacker with access to the algorithm output can learn about each individual record in the dataset. The original *pure* definition of DP has a privacy parameter ϵ (i.e., privacy budget), which bounds the probability that the attacker with access to the algorithm's output can determine whether a particular record was included in the dataset. DP has been extended to the notions of approximate DP, which includes a second parameter δ that is interpreted as the probability of information accidentally being leaked in addition to ϵ and Rényi DP [246].

DP has been widely adopted due to several useful properties: group privacy (i.e., the extension of the definition to two datasets that differ in k records), post-processing (i.e., privacy is preserved even after processing the output), and composition (i.e., privacy is composed if multiple computations are performed on the dataset). DP mechanisms for statistical computations include the Gaussian mechanism [113], the Laplace mechanism [113], and the Exponential mechanism [238]. The most widely used DP algorithm for training ML models is DP-SGD [1], and recent improvements include DP-FTRL [189] and DP matrix factorization [105].

By definition, DP provides mitigation against data reconstruction and membership inference attacks. In fact, the definition of DP immediately implies an upper bound on the success of an adversary in mounting a membership inference attack. Tight bounds on the success of membership inference have been derived by Thudi et al. [369]. However, DP does not provide guarantees against model extraction attacks, as this method is designed to protect the training data, not the model. Several papers have reported negative results after using differential privacy to protect against property inference attacks that aim to extract the properties of subpopulations in the training set [74, 233].

One of the main challenges of using DP in practice is setting up the privacy parameters to achieve a trade-off between the level of privacy and the achieved utility, which is typically measured in terms of accuracy for ML models. Analysis of privacy-preserving algorithms (e.g., DP-SGD) is often worst-case and not tight, and selecting privacy parameters based purely on theoretical analysis results in utility loss. Therefore, large privacy parameters are often used in practice (e.g., the 2020 U.S. Census release used $\epsilon = 19.61$), and the

exact privacy obtained in practice is difficult to estimate. Jagielski et al. [181] introduced *privacy auditing* with the goal of empirically measuring the actual privacy guarantees of an algorithm and determining privacy lower bounds by mounting privacy attacks. Many privacy auditing techniques are based on inserting *canaries* – synthetic and easy-to-identify out-of-distribution examples – into the training set, and then measuring the canary presence in model output. Auditing can also be performed with membership inference attacks [183, 427], but intentional insertion of strong canaries may result in better estimates of privacy leakage [181, 265]. Recent advances in privacy auditing include tighter bounds for the Gaussian mechanism [263] and rigorous statistical methods that allow for the use of multiple canaries to reduce the sample complexity of auditing [297]. Additionally, two efficient methods for privacy auditing with training a single model have been proposed: Steinke et al. [355] use multiple random data canaries without incurring the cost of group privacy; and Andrew et al. [10] used multiple random client canaries and cosine similarity test statistics to audit user-level private federated learning.

Differential privacy provides a rigorous notion of privacy and protects against membership inference and data reconstruction attacks. To achieve the best balance between privacy and utility, empirical privacy auditing is recommended to complement the theoretical analysis of private training algorithms.

There are other mitigation techniques against model extraction, such as limiting user queries to the model, detecting suspicious queries to the model, or creating more robust architectures to prevent side-channel attacks. However, these techniques can be circumvented by motivated and well-resourced attackers and should be used with caution. There are practice guides available for securing ML deployments [69, 274]. A completely different approach to potentially mitigating privacy leakage of a user's data is to perform **MACHINE UNLEARNING**, a technique that enables a user to request the removal of their data from a trained ML model. Existing techniques for machine unlearning are either exact (i.e., retraining the model from scratch or from a certain checkpoint) [45, 52] or approximate (i.e., updating the model parameters to remove the influence of the unlearned records) [139, 175, 268]. They offer different tradeoffs between computation and privacy guarantees, with exact unlearning methods offering stronger privacy, at additional computational cost.

3. Generative AI Taxonomy

GenAI is a branch of AI that develops models that can generate content (e.g., images, text, and other media) with similar properties to their training data. GenAI includes several different types of AI technologies with distinct origins, modeling approaches, and related properties, including: **GENERATIVE ADVERSARIAL NETWORKS**, **GENERATIVE PRE-TRAINED TRANSFORMER (GPT)**, and **DIFFUSION MODELS**, among others. Recently, GenAI systems have emerged with multi-modal content generation or comprehension capabilities [119], sometimes through combining two or more model types.

3.1. Attack Classification

While many attack types in the PredAI taxonomy apply to GenAI (e.g., data poisoning, model poisoning, and model extraction), recent work has also introduced novel AML attacks specific to GenAI systems.

Figure 2 shows a taxonomy of attacks in AML for GenAI systems. Similar to the PredAI taxonomy in Fig. 1, this taxonomy is first categorized by the system properties that attackers seek to compromise in each case, including **availability** breakdowns, **integrity** violations, and **privacy** compromises, as well as the additional category of AML attack relevant to GenAI of **misuse** enablement, in which attackers seek to circumvent restrictions placed on the outputs of GenAI systems (see Sec. 2.1.2). The capabilities that an adversary must leverage to achieve their objectives are shown in the outer layer of the objective circles. Attack classes are shown as callouts connected to the capabilities required to mount each attack. Where there are specific types of a more general class of attack (for example, a **jailbreak** is a specific kind of **direct prompting attack** attack), the specific attack is linked to the more general attack class through an additional callout. Certain attack classes are listed multiple times because the same attack technique can be used to achieve different attacker goals.

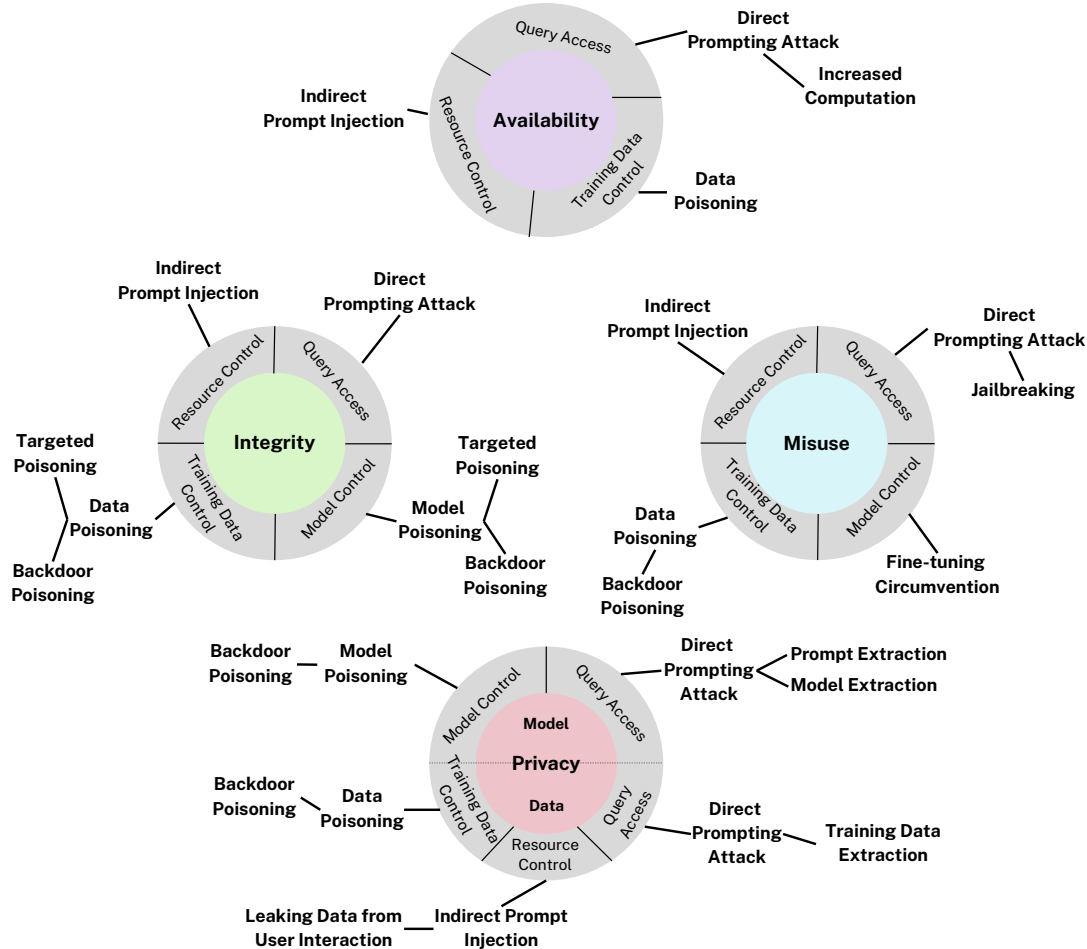


Figure 2. Taxonomy of attacks on GenAI systems

An attack can be further categorized by the learning stage to which it applies and by the attacker's knowledge and access. These are reviewed in the following sections. Where possible, the discussion broadly applies to GenAI models, though some attacks may be most relevant to particular kinds of GenAI models or model-based systems such as **RETRIEVAL-AUGMENTED GENERATION (RAG)** [RAG] systems, chatbots, or **AGENT** systems.

3.1.1. GenAI Stages of Learning

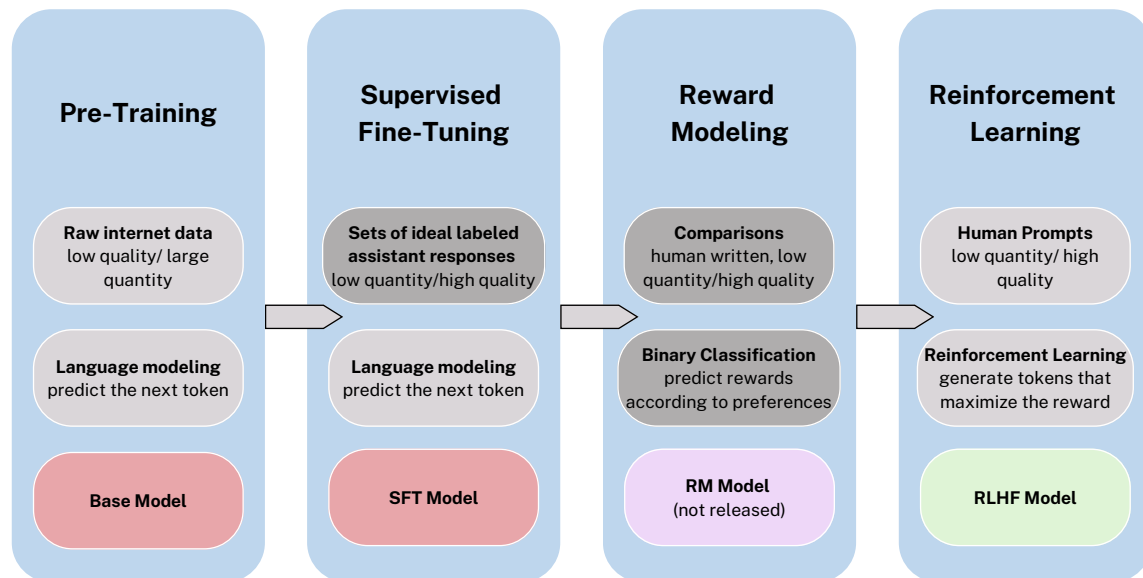


Figure 3. Example LLM Training Pipeline used for InstructGPT [281]

The GenAI development pipeline shapes the space of possible AML attacks against GenAI models and systems. In GenAI moreso than in PredAI, different activities such as data collection, model training, model deployment, and application development are often carried out by multiple different organizations or actors.

For example, a common paradigm in GenAI is the use of a smaller number of foundation models to support a diverse range of downstream applications. Foundation models are pre-trained on large-scale data using self-supervised learning in order to encode general patterns in text, images, or other data that may be relevant for many different applications [311]. Data at the scale used in foundation models is often collected from a variety of internet sources (which attackers can target, such as in DATA POISONING attacks).

This generalist learning paradigm equips foundation models with a variety of capabilities and tendencies — many of which are desirable, but some of which may be harmful or unwanted by the model developer. Techniques such as supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) can be used after initial pre-training to better align the base model with human preferences and to curb undesirable or harmful model outputs [281] (see Fig. 3). However, these interventions can later be targeted using AML techniques by attackers seeking to recover or re-enable potentially harmful capabilities.

Developers can make trained foundation models available to downstream users and developers in a variety of ways, including openly releasing the model's weights for re-use and

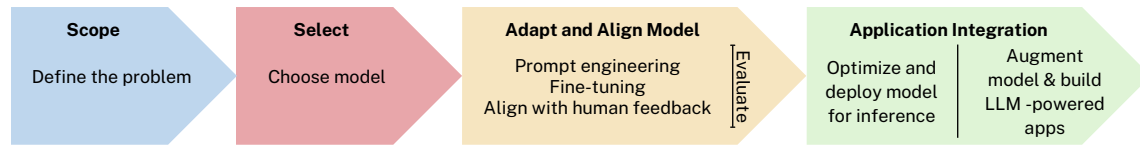


Figure 4. LLM enterprise adoption pipeline

modification, or hosting the model and offering access as a service through an API. These release decisions impact attacker capabilities that shape the space of possible AML attacks, such as whether attackers possess MODEL CONTROL.

Depending on how a foundation model has been made available, downstream developers can customize and build upon the model to create new applications, such as by further fine-tuning the model for a specific use case, or by integrating a foundation model with a software system, such as to build a retrieval-augmented generation (RAG) or agent (see Figure 4). Thus, a foundation model’s vulnerabilities to AML attacks can potentially impact a wide range of downstream applications and end users. At the same time, the specific application context in which a foundation model is integrated can create additional vectors for and risks from AML attacks, such as the potential exposure of application-specific data.

AML attacks differ and depend on different phases of the GenAI development lifecycle. One major division is between attacks that target the training stage and those that target model inference during the deployment stage.

Training-time attacks. [NISTAML.037] [Back to Index] The TRAINING STAGE for GenAI often consists of foundation model PRE-TRAINING and model FINE-TUNING. This pattern exists for generative image models, text models, audio models, and multimodal models, among others. Since foundation models are most effective when trained on large datasets, it has become common to scrape data from a wide range of public sources, increasing the vulnerability of these models to DATA POISONING attacks. Additionally, GenAI systems trained or fine-tuned by third parties are often used in downstream applications, leading to the risk of MODEL POISONING attacks from maliciously constructed models.

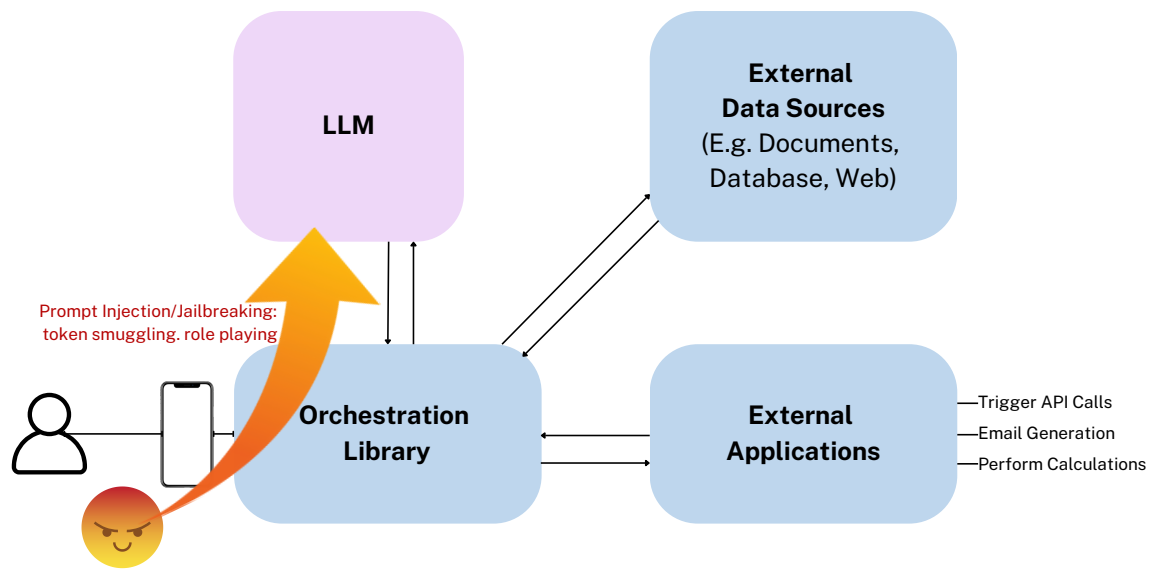


Figure 5. LLM enterprise adoption reference architecture

Inference-time attacks. The DEPLOYMENT STAGE for GenAI models and systems varies both based on how models are hosted or otherwise made available to users, and in how they are integrated into downstream applications. However, GenAI models and applications often share properties that leave them vulnerable to similar types of attacks. For example, underlying many of the security vulnerabilities in LLM applications is the fact that data and instructions are not provided in separate channels to the LLM, which allows attackers to use data channels to inject malicious instructions in inference-time attacks (a similar flaw to that which underlies decades-old SQL injection attacks). Many of the attacks in this stage are due to the following practices that are common in applications of text-based generative models:

1. **In-context instructions and system prompts:** [NISTAML.035] [Back to Index] The behavior of LLMs can be shaped through inference-time prompting, whereby the developer or user provides in-context instructions that are often prepended to the model's other input and context. These instructions comprise a natural language description of the model's application-specific use case (e.g., "You are a helpful financial assistant who responds gracefully and concisely...") and is known as a SYSTEM PROMPT. A PROMPT INJECTION overrides these instructions, exploiting the concatenation of untrusted user output to the system prompt to induce unintended behavior. For example, an attacker could inject a JAILBREAK that overrides the system prompt to cause the model to generate restricted or unsafe outputs. Since these prompts have been carefully crafted through prompt engineering and may be security-relevant, a PROMPT EXTRACTION attack may attempt to steal these system instructions. These attacks are also relevant to multimodal and text-to-image models.
2. **Runtime data ingestion from third-party sources:** In RETRIEVAL-AUGMENTED GENERA-

TION (RAG) applications, chatbots, and other applications in which GenAI models are used to interface with additional resources, context is often crafted at runtime in a query-dependent way and populated from external data sources (e.g., documents, web pages, etc.) that are to be used as part of the application. INDIRECT PROMPT INJECTION attacks depend on the attacker's ability to modify external sources of information that will be ingested into the model context, even if not provided directly by the primary system user.

3. **Output handling:** The output of an GenAI model may be used dynamically, such as to populate an element on a web page or to construct a command that is executed without any human supervision, which can lead to a range of availability, integrity or privacy violations in downstream applications if an attacker can induce behavior in this output that the developer has not accounted for.
4. **Agents:** An LLM-based AGENT relies on iteratively processing the *output* of an LLM (item 3 above) to perform a task and then provides the results as additional context back to the LLM input (item 2) [151, 155, 393]. For example, an agent system may select from among a configured set of external dependencies and invoke the code with templates filled out by the LLM using information in the context. Adversarial inputs into this context, such as from interactions with untrusted resources, could hijack the agent into performing adversary-specified actions instead, leading to potential security or safety violations.

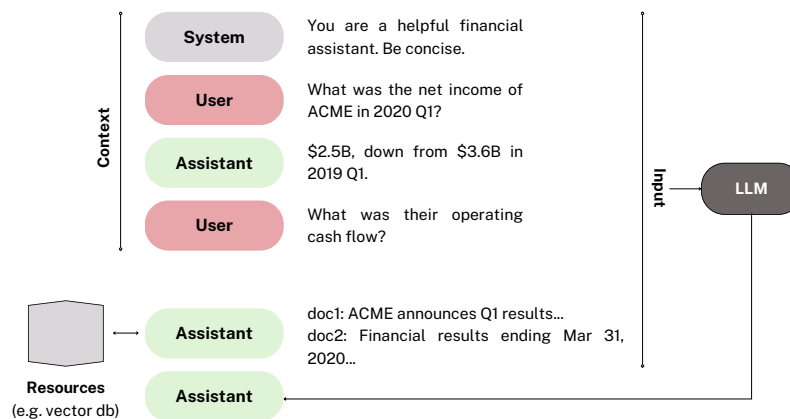


Figure 6. Retrieval-augmented generation

3.1.2. Attacker Goals and Objectives

As with PredAI, attacker objectives can be classified broadly along the dimensions of availability, integrity, and privacy, along with a new, GenAI-specific category of attacks designed to enable misuse.

- In an **AVAILABILITY BREAKDOWN attack** [NISTAML.01] [Back to Index], an attacker

seeks to interfere with a GenAI model or system to disrupt the ability of other users or processes to obtain timely and consistent access to its outputs or functionality.

- In an **INTEGRITY VIOLATION** attack [NISTAML.02] [Back to Index], an attacker seeks to interfere with a GenAI system to force it to misperform against its intended objectives and produce output that aligns with the attacker's objective. As users and businesses rely on GenAI systems to perform tasks like research and productivity assistance, these violations can allow attackers to weaponize the trust that these users place in GenAI systems.
- In a **PRIVACY COMPROMISE** attack [NISTAML.03] [Back to Index], an attacker seeks to gain unauthorized access to restricted or proprietary information that is part of a GenAI system, including information about a model's training data, weights or architecture; or sensitive information that the model accesses such as the knowledge base of a **RETRIEVAL-AUGMENTED GENERATION (RAG)** application. GenAI systems may be exposed to sensitive data (intentionally or otherwise) during training or inference, and attacks may seek to extract such information (e.g., through **INDIRECT PROMPT INJECTION** attacks, where a third party exfiltrates in-context user information [307], or a **MODEL EXTRACTION** attack to exfiltrate model information [61]).
- **Misuse enablement** [NISTAML.04] [Back to Index]. An additional attacker objective that is especially relevant in the GenAI context is the goal of **MISUSE ENABLEMENT**. In these attacks, an attacker seeks to deliberately circumvent technical restrictions imposed by the GenAI system's owner on its use, such as restrictions designed to prevent the system from producing outputs that could cause harm to others, cf. [325].

Technical restrictions refer in this context to defenses applied to the GenAI system such as the use of system prompts or RLHF for safety alignment. While the specific technical restrictions in place will vary between models, the techniques for circumventing such defenses are often common between different kinds of models and different kinds of misuse, allowing them to be taxonomized as a part of AML without specificity as to the particular kinds of misuse that model developers seek to prevent.

3.1.3. Attacker Capabilities

AML attacks can be taxonomized with respect to the capabilities that an attacker has in controlling inputs to the GenAI model or system. These capabilities include:

- **TRAINING DATA CONTROL**: The attacker might take control of a subset of the training data by inserting or modifying training samples. This capability is used in **DATA POISONING** attacks.
- **QUERY ACCESS**: Many GenAI models and their applications are deployed as services that can be accessed over the internet by users. In these cases, attackers can sub-

mit adversarially crafted queries to the model to elicit specific desired behavior or extract information. This capability is used for **PROMPT INJECTION**, **PROMPT EXTRACTION**, and **MODEL EXTRACTION** attacks. Query access can vary based on the degree of generation control (e.g., modifying the temperature or adding a logit bias) and the richness of the returned generation (e.g., with or without log probabilities or multiple choices).

- **RESOURCE CONTROL**: The attacker might modify resources (e.g., documents, web pages) that will be ingested by the GenAI model at runtime. This capability is used for **INDIRECT PROMPT INJECTION** attacks.
- **MODEL CONTROL**: The attacker might have the ability to modify model parameters, such as through public fine-tuning APIs or openly accessible model weights. This capability is used in **MODEL POISONING** attacks, as well **FINE-TUNING CIRCUMVENTION** attacks which remove refusal behavior or other model-level safety interventions [132, 153, 300].

As in PredAI, attackers can also vary in their knowledge of the underlying ML model, from full knowledge of the ML system including model weights (**white-box attacks**), to minimal knowledge and systems with deliberately obscured or misleading information (**black-box attacks**), to somewhere in between (**gray-box attacks**). See Sec. 2.1.4, which discusses attacker knowledge in greater detail and applies to GenAI attacks.

3.2. Supply Chain Attacks and Mitigations

[NISTAML.05] [\[Back to Index\]](#)

Since AI is software, it inherits many of the vulnerabilities of the traditional software supply chain, such as reliance on third-party dependencies. AI development also introduces new types of dependencies, including data collection and scoring, the integration or adaptation of third-party-developed AI models, and the integration of third-party-developed plugins into AI systems. Mitigating the security challenges in AI supply chain management is complex and requires a multifaceted approach that combines existing practices for software supply chain risk management with the management of AI-specific supply chain risks, such as through the use of provenance information for the additional artifacts involved [159, 267]. Studies of real-world security vulnerabilities against ML suggest that security is best addressed comprehensively and by considering the full attack surface, including data and model supply chains, software, and network and storage systems [17, 370]. While all of these supply chain risks are critical in the broader context of securing AI systems, there are certain types of attacks that rely on exploiting the specific statistical and data-based properties of ML systems, thus falling within the domain of AML.

3.2.1. Data Poisoning Attacks

The performance of GenAI text-to-image and text-to-text models has been found to scale with dataset size (among other properties like model size and data quality); for example, Hoffmann et al. [161] suggest compute-optimally training a 520 billion parameter model may require 11 trillion tokens of training data. Thus, it has become common for GenAI foundation model developers to scrape data from a wide range of sources. In turn, the scale of this data and the diversity of its sources provides a large potential attack surface into which attackers may seek to insert adversarially constructed data points. For example, dataset publishers may provide a list of URLs to constitute a training dataset, and attackers may be able to purchase some of the domains that serve those URLs and replace the site content with their own malicious content [57].

Beyond the vast quantities of pre-training data, data poisoning attacks may also affect other stages of the LLM training pipeline, including instruction tuning [389] and reinforcement learning from human feedback [305], which may intentionally source data from a large number of human participants.

As with PredAI models (see Sec. 2.1), data poisoning attacks could lead to attackers controlling model behavior through the insertion of a backdoor (see BACKDOOR POISONING ATTACK) such as a word or phrase that, when submitted to a model, acts as a universal JAILBREAK [305]. Attackers could also use data poisoning attacks to modify model behavior on particular user queries (see TARGETED POISONING ATTACK), such as causing the model to incorrectly summarize or otherwise produce degenerate outputs in response to queries that contain a particular trigger word or phrase [389]. These attacks may be practical—requiring a relatively small portion of the total dataset [46]—and may lead to a range of bad outcomes, such as code suggestion models which intentionally suggest insecure code [3].

3.2.2. Model Poisoning Attacks

[NISTAML.051] [Back to Index]

In GenAI, it is common for developers to use foundation models developed by third parties. Attackers can take advantage of this fact by offering maliciously designed models, such as pre-trained models that enable a BACKDOOR POISONING ATTACK or TARGETED POISONING ATTACK. While this attack relies on the attacker having control over the initial poisoned model, researchers have identified attacks in which malicious backdoors in pre-trained models can persist even after downstream users fine-tune the model for their own use [201] or apply additional safety training measures [170].

3.2.3. Mitigations

GenAI poisoning mitigations largely overlap with PredAI poisoning mitigations (see Sec. 2.3). For preventing data poisoning with web-scale data dependencies, this includes ver-

ifying web downloads as a basic integrity check to ensure that domain hijacking has not injected new sources of data into the training dataset [57]. That is, the provider publishes cryptographic hashes, and the downloader verifies the training data. Data filtering can also attempt to remove poisoned samples, though detecting poisoned data within a large training corpus may be very difficult.

While traditional software supply chain risk management practices such as vulnerability scanning of model artifacts can help manage some kinds of AI supply chain risks, new approaches are required to detect vulnerabilities in models such as those introduced through model poisoning attacks. Current proposed approaches include using methods from the field of mechanistic interpretability to identify backdoor features [67] and detecting and counteracting triggers when they are seen at inference time. Beyond these mitigations, risks can be reduced by understanding models as untrusted system components and designing applications such that risks from attacker-controlled model outputs are reduced [266].

3.3. Direct Prompting Attacks and Mitigations

[NISTAML.018] [Back to Index] **DIRECT PROMPTING ATTACK** attacks arise when the attacker is the primary user of the system, interacting with the model through query access. A subset of these attacks, in which the main user provides in-context instructions that are appended to higher-trust instructions like those provided by the application designer (such as the model's **SYSTEM PROMPT**), are known as **DIRECT PROMPT INJECTION** attacks.

As in PredAI, attacks may be applicable to a single setting and model, or may instead be universal (affecting models on a range of separate queries, see Sec. 2.2.1) and/or transferable (affecting models beyond the model they are found on, see Sec. 2.2.3).

An attacker may have a variety of goals when performing these attacks [219, 220, 337], such as to:

- **Enable misuse.** Attackers may use direct prompting attacks to bypass model-level defenses that a model developer or deployer has created to restrict models from producing harmful or undesirable output [237]. A **JAILBREAK** is a direct prompting attack intended to circumvent restrictions placed on model outputs, such as circumventing refusal behavior to enable misuse.
- **Invade privacy.** Attackers may use direct prompting to extract the system prompt or reveal private information that was provided to the model in context but not intended for unfiltered access by the user.
- **Violate integrity.** When LLMs are used as agents, an attacker may use direct prompting attacks to manipulate tool usage and API calls, and potentially compromise the backend of the system (e.g. executing attacker's SQL queries).

3.3.1. Attack Techniques

A range of techniques exist for launching direct prompting attacks, many of which generalise across various attacker objectives. With a focus on direct prompting attacks to enable misuse, we note the following broad categories of direct prompting techniques (see [393]):

- **Optimization-based attacks** design attack objective functions and use gradient or other search-based methods to learn adversarial inputs that cause a particular behavior, similar to PredAI attacks discussed in Sec. 2.2.1. Objective functions may be designed to force affirmative starts (e.g., looking for responses that begin with “Sure”, which may indicate compliance with a malicious request [60, 320, 448]) or other metrics of attack success (e.g., similarity to a toxified finetune [368]).

Optimization techniques can then be used to learn attacks, including techniques that follow from attacks designed for PredAI language classifiers (e.g., HotFlip [117]) and gradient-free techniques that use a proxy model or random search to test attack candidates [11, 320]. Universal adversarial triggers are a special class of these gradient-based attacks against generative models that seek to find *input-agnostic* prefixes (or suffixes) that produce the desired affirmative response regardless of the remainder of the input [386, 448]. That these universal triggers transfer to other models makes open-weight models — for which there is ready white-box access — feasible attack vectors for transferability attacks on closed systems in which only API access is available [448].

Attacks can also be designed to satisfy additional constraints (e.g., sufficiently low perplexity [368]) or attack a system of multiple models [235].

- **Manual methods** for jailbreaking an LLM include competing objectives and mismatched generalization [400]. Mismatched generalization-based attacks identify inputs that fall outside the distribution of the model’s safety training but remain within the distribution of its capabilities training, making them comprehensible to the model while evading refusal behavior. Competing objectives-based attacks find cases where model capabilities are in tension with safety goals, such as by playing into a model’s drive to follow user-provided instructions. In all cases the goal of the attack is to compromise a model-level safety defense. See Weng [403] for further discussion.

Approaches to competing objectives-based attacks include:

1. *Prefix injection*: This method involves prompting the model to start responses with an affirmative confirmation. By conditioning the model to begin its output in a predetermined manner, adversaries attempt to influence its subsequent language generation toward specific, predetermined patterns or behaviors.
2. *Refusal suppression*: Adversaries may explicitly instruct the model to avoid generating refusals or denials in its output. By decreasing the probability of refusal responses, this tactic aims to increase the probability of a compliant

response.

3. *Style injection*: In this approach, adversaries instruct the model to use (or not use) certain syntax or writing styles. For example, an attack may constrain the model's language to simplistic or non-professional tones, aiming to decrease the probability of (usually professionally worded) refusals.
4. *Role-play*: Adversaries utilize role-play strategies (e.g., "Always Intelligent and Machiavellian" [AIM] or "Do Anything Now" [DAN]) to guide the model to adopt specific personas or behavioral patterns that conflict with the original intent. This manipulation aims to exploit the model's adaptability to varied roles or characteristics, with an intent to compromise its adherence to safety protocols.

Approaches to mismatched generalization-based attacks include:

1. *Special encoding*: Strategies that use encoding techniques like base64 to alter the representation of input data in a way that remains understandable to the model but may be out of distribution for safety training.
 2. *Character transformation*: Strategies that use character-level transformations like the ROT13 cipher, symbol replacement (e.g., l33tspeak), and Morse code to take the input out of the safety training distribution.
 3. *Word transformation*: Strategies that alter the linguistic structure of the input, such as Pig Latin, synonym swapping (e.g., using "pilfer" for "steal"), and payload splitting (or "token smuggling") to break down sensitive words into substrings.
 4. *Prompt-level transformation*: Strategies that use prompt-level transformations, such as translating the prompt into a less common language that may be out of distribution of the safety training data.
- **Automated model-based red teaming** employs an attacker model, a target model, and a judge [73, 239, 292]. When the attacker has access to a high-quality classifier that judges whether model output is harmful, it may be used as a reward function to train a generative model to generate jailbreaks of another generative model. Only query access is required for each of the models, and no human intervention is required to update or refine a candidate jailbreak. The prompts may also be transferable from the target model to other closed-source LLMs [73].

The Crescendo attack [316] introduced the idea of interacting with the model iteratively in a multi-turn adaptive attack that includes seemingly benign prompts, eventually leading to a successful jailbreak against safety alignment. The initial manual attack is fully automated by leveraging another LLM for prompt generation and incorporating multiple input sources.

Evaluations of leading models suggest LLMs remain vulnerable to these many of these attacks [11, 338, 381].

3.3.2. Information Extraction

[NISTAML.038] [Back to Index] Both during training and at run-time, GenAI models are exposed to a range of information which may be of interest to attackers, like personally identifying information (PII) in the training data, sensitive information in RETRIEVAL-AUGMENTED GENERATION (RAG) databases provided in-context, or even the SYSTEM PROMPT constructed by the application designer. Additionally, features of the model itself—such as the model weights or architecture—may be targets of attack. Though many of the techniques in Sec. 3.3.1 apply to extracting such data, we note several specific goals and techniques specific to data extraction.

Leaking sensitive training data. Carlini et al. [59] were the first to practically demonstrate TRAINING DATA EXTRACTION attacks in generative language models. By inserting canaries—synthetic, easy-to-recognize out-of-distribution examples—in the training data, they developed a methodology for extracting the canaries and introduced a metric called *exposure* to measure memorization. Subsequent work demonstrated the risk of data extraction in LLMs based on transformers (e.g., GPT-2 [63]) by prompting the model with different prefixes and mounting a membership inference attack to determine which generated content was part of the training set. Since these decoder stack transformers are autoregressive models, a verbatim textual prefix about personal information can sometimes result in the model completing the text input with sensitive information that includes email addresses, phone numbers, and locations [229]. This behavior of verbatim memorization of sensitive information in GenAI language models has also been observed in more recent transformer models with the additional characterization of extraction methods [165]. Unlike PredAI models in which tools like Text Revealer are created to reconstruct text from transformer-based text classifiers [434], GenAI models can sometimes simply be asked to repeat private information that exists in the context as part of the conversation. Results show that information like email addresses can be revealed at rates exceeding 8% for certain models. However, their responses may wrongly assign the owner of the information and be otherwise unreliable. In general, extraction attacks are more successful when the model is seeded with more specific and complete information — the more the attacker knows, the more they can extract. Researchers have leveraged this fact to incrementally extract fragments of copyrighted New York Times articles from LLMs by seeding it with a single sentence, and allowing the LLM to recurrently extract additional text [356]. Intuitively, larger models with a higher capacity are more susceptible to exact reconstruction [56]. Fine-tuning interfaces also amplify the risk of data extraction attacks, as demonstrated by an attack that extracts PII from pre-training data using fine-tuning API for open-weight models [83], though this is not a direct prompting attack.

Prompt and context stealing. Prompts are vital to align LLMs to a specific use case and are

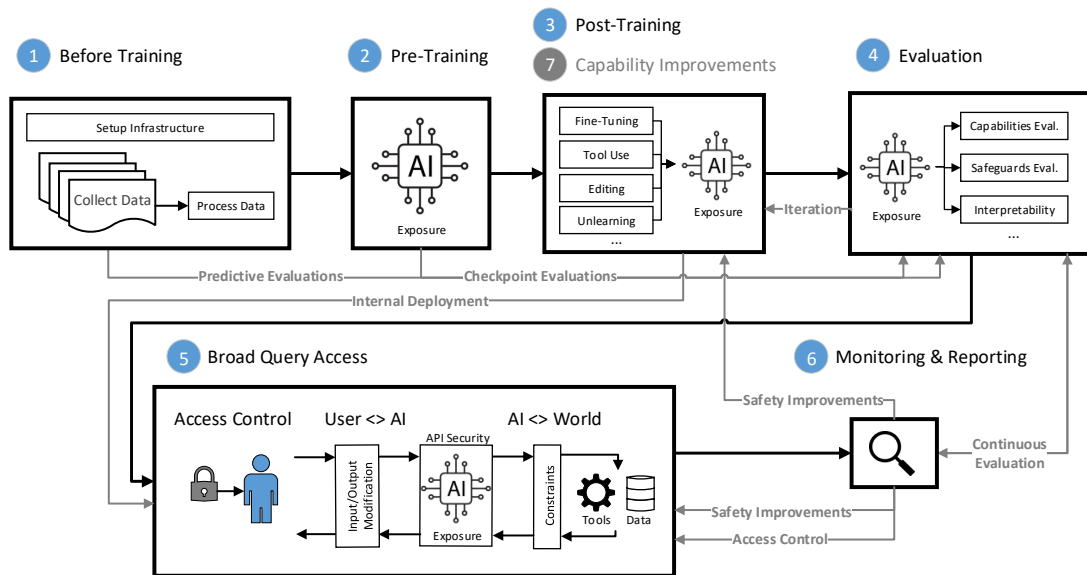


Figure 7. Map of the development and deployment life cycle of an AI model for broad-scale query access

a key ingredient to their utility in following human instructions. These prompts can therefore be regarded as commercial secrets, and are sometimes the target of direct prompting attacks. PromptStealer is a learning-based method that reconstructs prompts from text-to-image models using an image captioning model and a multi-label classifier to steal both the subject and the prompt modifiers [339]. For certain LLMs, researchers have found that a small set of fixed attack queries (e.g., Repeat all sentences in our conversation) were sufficient to extract more than 60 % of prompts across certain model and dataset pairs [439]. In some cases, effective prompts may draw from significant technical or domain expertise; prompt-stealing attacks may violate or threaten these investments. Furthermore, in RAG applications (see Fig. 6), the same techniques can be used to extract sensitive information provided in the LLMs' context. For example, rows from a database or text from a PDF document that are intended to be summarized generically by the LLM can be verbosely extracted by simply asking for them via direct prompting, or performing simple prompting attacks.

Model extraction. As in PredAI (Sec. 2.4.4), attackers may perform **MODEL EXTRACTION** attacks which attempt to learn information about the model architecture and parameters by submitting specially-crafted queries. Recently, Carlini et al. [61] demonstrated that such information could be extracted from black-box production LLMs, deriving previously unknown hidden dimensions and the embedding projection layer (up to symmetries).

3.3.3. Mitigations

The following defense strategies can be employed throughout the deployment life cycle of an AI model or system to reduce the risk that the model or system will be vulnerable to direct prompt injections. The numbers in parentheses refer to the numbering in Fig. 7, which shows a map of the deployment life cycle for broad-scale query access.

- **Interventions during pre-training (2) and post-training (3).** A range of training strategies have been proposed to increase the difficulty of accessing harmful model capabilities through direct prompt injection, including safety training during pre-training [197] or post-training [147, 445], adversarial training methods [340], and other methods to make jailbreak attacks more difficult [447].
- **Interventions during evaluation (4).** Evaluations can measure the vulnerability of models to query-based attacks, which can then inform trust and affordance decisions, as well as developer and user education. Evaluations can include broad automated vulnerability assessments [72, 107, 324] as well as targeted expert red teaming [381] and bug bounties [16]. Current evaluation approaches, though a useful tool, may underestimate vulnerabilities accessible to actors with more time, resourcing, or luck. Evaluations measure model vulnerabilities at a particular moment in time; assessments may change if new attacks are developed, additional data is collected post-training, or model capabilities are improved. Continuous evaluations following deployment can help combat these challenges.
- **Interventions during deployment (5).** A broad set of deployment-time interventions have been proposed:
 - *Prompt instruction and formatting techniques.* Model instructions can cue the model to treat user input carefully, such as by wrapping user input in XML tags, appending specific instructions to the prompt, or otherwise attempting to clearly separate system instructions from user prompts [14, 206, 219].
 - *Detecting and terminating harmful interactions.* Rather than preventing harmful model generations, AI systems may be able to detect these generations and terminate interactions. Several open [5, 6, 154] and closed [18, 204, 313] solutions have explored LLM-based detection systems with distinctly prompted and/or fine-tuned models that classify user input and/or model output as harmful or undesirable. These may provide supplementary assurance through a defense-in-depth philosophy. However, these detection systems are also vulnerable to attacks [235] and may have correlated failures to the main models that they are monitoring. Some lines of research have investigated constraining the space of generations to enable deterministic guardrails [306]. Early work suggests that interpretability-based techniques can also be used to detect anomalous input [31], as well as keyword- or perplexity-based defenses [9, 164].

- *Prompt stealing detection.* A common approach to mitigating prompt stealing is to compare the model utterance to the prompt, which is known by the system provider. Defenses differ in how this comparison is made, which might include looking for a specific token, word, or phrase, as popularized by [59], or comparing the n-grams of the output to the input [439]. Similarly, defenses for prompt stealing have yet to be proven rigorous.
- *Input modification.* User input can additionally be modified prior to being passed to the model, such as paraphrasing or retokenizing [182]. However, such methods may be expensive and/or have trade-offs with model performance.
- *Aggregating output from multiple prompts.* Motivated by randomized smoothing [95] used to improve robustness against evasion attacks for ML classifiers, SmoothLLM [312] proposes aggregating the LLM output from multiple randomly perturbed prompts. This defense incurs a cost of generating multiple LLM queries for each prompt and might reduce the quality of the generated output.
- *Monitoring and response.* Following deployment, monitoring and logging of user activity may allow model deployers to identify and respond to instances of attempted and successful direct prompt injection attacks [266]. This response could include banning or otherwise acting against users if their intentions appear malicious, or remediating the prompt injection vulnerability in the event of a successful attack. Standard user- or organization-level vetting or identity verification procedures, as well as clear incentive mechanisms (such as a policy of restricting model access in response to violations) may enhance the efficacy of this mitigation.
- *Usage restrictions.* Other interventions have focused on choices about how models are offered to users: for example, the efficacy of some attacks can be reduced by limiting the inference parameters that are accessible to users (e.g., temperature or logit bias), as well as the richness of the model generations returned (e.g., logit probabilities) [250]. Additionally, limiting the release of public information [252, 266] and artifacts [249] and restricting the total number of model queries available to users [251] may make attacks more challenging. These techniques may have additional drawbacks in limiting positive use cases.

Indirect mitigations. Despite the growing number of proposed defenses at both the model and the system levels, recent findings suggest that current generation models remain highly vulnerable to direct prompt injection attacks [11, 381]. Thus, other potential mitigations for prompt injection rely not on directly increasing an AI system’s robustness against such attacks, but instead on designing systems under the assumption that the AI model can and will produce malicious output if it is exposed to malicious actors. For example, deployers can design AI systems under the assumption that models with access to sensitive data

or the ability to take undesirable actions may leak that data or take those actions [266]. Additionally, developers or deployers might use other technical mitigations to reduce the misuse potential of outputs obtained through direct prompt injection, such as:

- *Training data sanitization.* Model training data can be sanitized to remove sensitive or toxic content and data that are largely or exclusively relevant for developing undesirable capabilities. Such sanitization may prevent harmful capabilities from being learned and reduce the potential harms from direct prompt injection, though they may harm generalization and harmful content detection abilities [224].
- *Unlearning.* There have also been attempts to “unlearn” harmful knowledge or capabilities post-training [212], with the goal of reducing harms from maliciously directed models [158]. However, these methods remain vulnerable to adversarial attacks, including attacks on jailbreak-specific training approaches [367] and inversion attacks on unlearning methods [328], which extract supposedly unlearned data.
- *Watermarking.* Developers or deployers may watermark content generated by an AI model to help trace its provenance, distinguish it from human-generated content, and reduce risks from malicious use cases (e.g., by flagging content as model-generated when it appears online). While the literature has proposed various techniques with different strengths and weaknesses [194], there is no watermarking technique that is universally effective and robust under all circumstances. Many powerful attacks have been developed against watermarking with high success rates [188, 319]. Moreover, theoretical impossibility results regarding the robustness of watermarking have also been established [432].

Finally, beyond interventions at the developer or deployer levels, society and infrastructure can become more resilient to maliciously directed model capabilities over time [7, 33]. For example, defenders could adopt AI-based vulnerability discovery tools [99] to make their systems more resilient to malicious actors misusing GenAI models to find vulnerabilities for exploitation.

3.4. Indirect Prompt Injection Attacks and Mitigations

[NISTAML.015] [Back to Index] Many use cases for GenAI models involve models interacting with additional resources, from an internet-connected AGENT to a RETRIEVAL-AUGMENTED GENERATION (RAG) system depicted in Fig. 6. Because GenAI models combine the *data* and *instruction* channels, attackers can leverage the data channel to affect system operations by manipulating resources with which the system interacts. Thus, INDIRECT PROMPT INJECTION attacks are enabled by RESOURCE CONTROL that allows an attacker to indirectly (or remotely) inject system prompts without directly interacting with the application [146, 408]. Indirect prompt injection attacks can result in violations across at least three categories of attacker goals: 1) availability violation, 2) integrity violation, and 3) privacy compromise. However, unlike in direct prompt injection attacks, indirect prompt injection attacks are mounted not

by the primary user of a model but instead by a third party. In fact, in many cases, it is the primary user of the model who is harmed by the compromise of the integrity, availability, or privacy of the GenAI system through an indirect prompt injection attack.

3.4.1. Availability Attacks

[NISTAML.016] [\[Back to Index\]](#) Attackers can manipulate resources to inject prompts into GenAI models that are designed to disrupt the availability of the model for legitimate users. Availability attacks can indiscriminately render a model unusable (e.g., failure to generate helpful outputs) or specifically block certain capabilities (e.g., specific APIs) [146].

Attacker techniques. Researchers have demonstrated several proof-of-concept methods by which attackers can disrupt the availability of a GenAI system:

- **Time-consuming background tasks.** [NISTAML.017] [\[Back to Index\]](#) An indirectly injected prompt can instruct the model to perform a time-consuming task prior to answering the request. The prompt itself can be brief, such as by requesting looping behavior in the evaluating model [146].
- **Inhibiting capabilities.** An indirectly injected prompt can instruct the model that it is not permitted to use certain APIs (e.g., the search API for an internet-connected chatbot). This selectively disarms key components of the service [146].
- **Disruptive output formatting.** An attacker can use indirect prompt injection to instruct the model to modify its output in a way that disrupts the availability of the system. For example, an attacker could instruct the model to replace the characters in retrieved text with homoglyph equivalents, disrupting subsequent API calls [146]; or could request that the model begins each sentence with an `<|endoftext|>` token, forcing the model to return an empty output [146].

3.4.2. Integrity Attacks

[NISTAML.027] [\[Back to Index\]](#)

Through indirect prompt injection, attackers can use malicious resources to prompt GenAI systems to become untrustworthy and generate content that deviates from benign behavior to align with adversarial objectives. These attacks often involve disrupting the model's behavior in subtle ways that may not be obvious to the end user.

For example, researchers have demonstrated attacks through indirect prompt injection that can cause a GenAI system to produce arbitrarily incorrect summaries of sources, to respond with attacker-specified information, or to suppress or hide certain information sources [146]. Attackers could use these capabilities to weaponize GenAI systems such as internet-connected chatbots against their users for a range of malign purposes, including spreading targeted misleading information, recommending fraudulent products or services, or redirecting consumers to malicious websites that spoof legitimate log-in pages or

contain downloadable malware. Attackers may also use indirect prompt injection attacks to hijack a GenAI **AGENT**, causing it to perform a malicious, attacker-specified task instead of (or in addition to) its intended, user-provided task [353].

Attacker techniques. Researchers have demonstrated integrity attacks through malicious resources that manipulate the primary task of the LLM:

- **Jailbreaking.** Attackers can leverage techniques for indirect prompt injection that are similar to those used in direct prompt injection attacks, such as using a **TAILBREAK** that allows the attacker to substitute their own malicious instructions in place of the model's **SYSTEM PROMPT**. As in direct prompting attacks, these attacks may be crafted through optimization-based or manual methods, and may rely on techniques such as mismatched generalization.
- **Execution triggers.** Researchers have automated manual indirect prompt injection attacks using execution triggers generated via optimization with a technique called Neural Exec [287]. These execution triggers can also persist through RAG processing pipelines that include multiple phases, such as chunking and contextual filtering.
- **Knowledge base poisoning.** The knowledge database of a RAG system can be poisoned to achieved targeted LLM output to specific user queries, as in PoisonedRAG [449]. Recently, a general optimization framework called Phantom [75] has shown how a single poisoned document can be crafted and inserted into the knowledge database of a RAG system to induce a number of adversarial objectives in the LLM generator.
- **Injection hiding.** Attackers may use techniques to hide or obfuscate their injections, such as by hiding injections in non-visible portions of a resource; using multi-stage injections, in which the initial injection directs the model to visit another resource which contains additional injections; or encoding injection commands such as in Base64 and then instructing the model to decode the sequence [146].
- **Self-propagating injections.** Attackers may be able to use indirect prompt injection attacks to turn GenAI systems into vectors for spreading attacks. For example, an attacker could send a malicious email that, when read by a model integrated as part of an email client, instructs the model to spread the infection by sending similar malicious emails to everyone in the user's contact list. In this way, certain malicious prompts could serve as *worms* [146].

3.4.3. Privacy Compromise

Attackers can use indirect prompt injection attacks to compromise the privacy of a GenAI system or its primary users. For example, attackers could use indirect prompt injection attacks to compel a model to leak information from restricted resources, such as a user's private data that is processed by the GenAI system. Alternately, in a blend of integrity and

privacy attacks, an attacker could gather information about the primary user or users of the system by instructing a model to obtain and then leak that information.

Attacker techniques. Researchers have theorized and demonstrated a variety of indirect prompt injection attacks to compromise information from internet-connected chatbots, RAG systems, and other GenAI systems. Some of these techniques include:

- **Compromising connected resources.** [NISTAML.039] [\[Back to Index\]](#) Attackers can use prompt injection attacks to cause a GenAI system to leak private information from the restricted resources it can access. For example, a model integrated as part of an email client could be prompted to forward certain emails to an attacker-controlled inbox [146]. Researchers have identified injection attacks that can force a model to exfiltrate user-uploaded data by querying an attacker-controlled URL with the sensitive data [298].
- **Leaking information from user interactions.** [NISTAML.036] [\[Back to Index\]](#) Researchers have demonstrated a proof-of-concept indirect prompt injection attack in which they inject instructions for a model to persuade the end user to reveal a piece of information (in this case, their name) that the model then leaks to the attacker, such as by directly querying an attacker-controlled URL with the information or suggesting such a URL to the user to visit [146]. Attackers may also be able to exploit features like markdown image rendering to exfiltrate data [323].

3.4.4. Mitigations

Various techniques (see Sec. 3.3.3) can be used throughout the development and deployment life cycle (Fig. 7) to mitigate attacks, including:

- Several training techniques have been developed to mitigate against indirect prompt injection, including fine-tuning task-specific models [296] and training models to follow hierarchical trust relationships in prompts [387].
- Detection schemes have been proposed to detect indirect prompt injection, and many LLM-based defenses have been designed to mitigate both direct and indirect prompt injection [6, 18, 154, 204, 313].
- A range of input processing methods have been proposed to combat indirect prompt injection, including filtering out instructions from third-party data sources [146], designing prompts to help aid LLMs in separating trusted and untrusted data (i.e., spotlighting [160, 206]), or instructing models to disregard instructions in untrusted data [206].

Many of the defenses described in the context of direct prompt injection can also be adapted to mitigate indirect prompt injection. Because current mitigations do not offer full protection against all attacker techniques, application designers may design systems

with the assumption that prompt injection attacks are possible if a model is exposed to untrusted input sources, such as by using multiple LLMs with different permissions [145, 405] or by allowing models to interact with potentially untrustworthy data sources only through well-defined interfaces [410]. Additionally, public education efforts can inform model users and application designers of the risks of indirect prompt injection [266].

3.5. Security of Agents

An increasingly common use of GenAI models is constructing an (often LLM-based) **AGENT**, a software system that iteratively prompts a model, process its outputs – such as to select and call a function with specified inputs – and provides the results back to the model as a part of its next prompt [151, 155, 393]. Agents may be equipped to use tools such as web-browsing or code interpreters, and may have additional features such as memory and/or planning capabilities.

Because agents rely on GenAI systems to plan and execute their actions, they can be vulnerable to the many of the above categories of attacks against GenAI systems, including direct and indirect prompt injection. However, because agents can take actions using tools, these attacks can create additional risks in this context, such as enabling actors to hijack agents to execute arbitrary code or exfiltrate data from the environment in which they are operating. Security research focused specifically on agents is still in its early stages, but researchers have begun to evaluate the vulnerability of agents to particular AML attacks [12, 430] and to propose interventions to manage the security risks posed by agents [24].

3.6. Benchmarks for AML Vulnerabilities

There are several publicly available benchmarks for evaluating models' vulnerability to AML attacks. Datasets like JailbreakBench [72], AdvBench [448], HarmBench [237], StrongREJECT [351], AgentHarm [12], and Do-Not-Answer [399] provide benchmarks for evaluating models' susceptibility to jailbreaks. TrustLLM [169] is a benchmark intended to evaluate six dimensions of trust in LLMs: truthfulness, safety, fairness, robustness, privacy, and machine ethics. AgentDojo [101] is an evaluation framework for measuring the vulnerability of AI agents to prompt injection attacks in which the data returned by external tools hijacks the agent to execute malicious tasks. Additionally, open-source tools like Garak [106] and PyRIT [364] are intended to help developers identify vulnerabilities to AML attacks in models. Finally, several unlearning benchmarks have recently been proposed [212, 234].

4. Key Challenges and Discussion

4.1. Key Challenges in AML

There are several fundamental challenges that make fully addressing the problems discussed in this report more difficult. We discuss several here: The trade off between increasing accuracy (or average-case performance) and other attributes including robustness (or worst-case performance); the theoretical limitations and results that imply that fully robust systems may be mathematically impossible without additional assumptions; and the challenge of evaluating progress in AML mitigations rigorously and robustly.

4.1.1. Trade-Offs Between the Attributes of Trustworthy AI

The trustworthiness of an AI system depends on all of the attributes that characterize it [274]. There are trade-offs between explainability and adversarial robustness [176, 245] and between privacy and fairness [178]. For example, AI systems that are optimized for accuracy alone tend to underperform in terms of adversarial robustness and fairness [71, 111, 302, 379, 433]. Conversely, an AI system that is optimized for adversarial robustness may exhibit lower accuracy and deteriorated fairness outcomes [32, 391, 433]. Unfortunately, it may not be possible to simultaneously maximize the performance of an AI system with respect to these attributes.

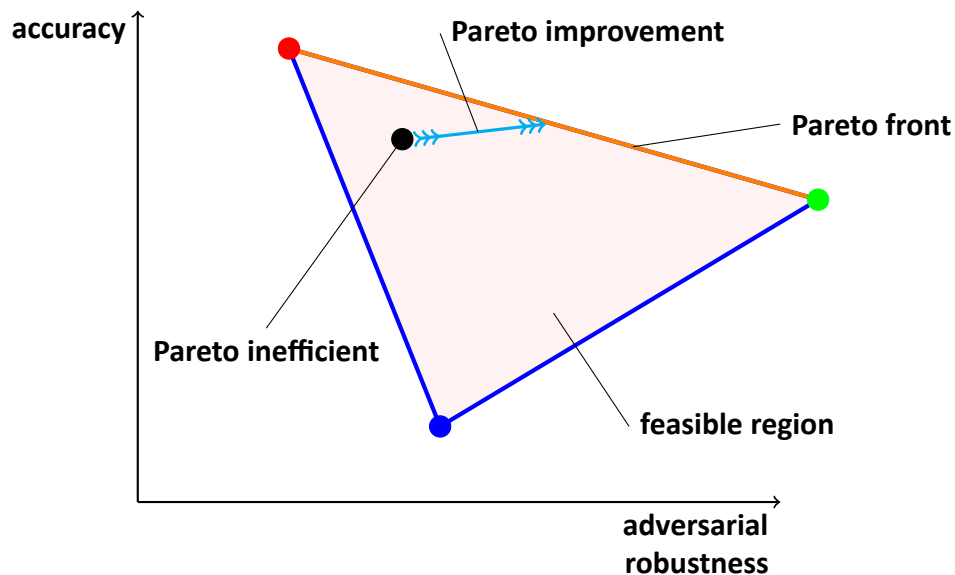


Figure 8. Pareto optimality

The full characterization of the trade-offs between the different attributes of trustworthy AI is an open research problem that is gaining importance with the adoption of AI technology in many areas of modern life. One promising practical approach is based on the concept of

multi-objective optimization and Pareto optimality [285, 286]. In most cases, there is no mathematically best trade-off. However, Fig. 8 illustrates a hypothetical example of a trade-off between accuracy and adversarial robustness. Any point in the feasible region that is not on the Pareto front is a bad point (i.e., Pareto inefficient). There is a better solution (i.e., Pareto improvement) that can significantly help with one objective without harming the other, which is a goal of Pareto optimization. Moreover, if there is a single optimum in some use case, Pareto optimization naturally attains it. Organizations may need to accept trade-offs between these properties and decide which of them to prioritize depending on the AI system, the use case, and other relevant implications of the AI technology [274, 326, 382].

4.1.2. Theoretical Limitations on Adversarial Robustness

Given the multitude of powerful attacks, appropriate mitigations must be designed before AI systems are deployed in critical domains. This challenge is exacerbated by the lack of theoretically secure ML algorithms for many tasks in the field (see Sec. 1). This implies that designing mitigations is an inherently ad hoc and fallible process, though there are practice guides for securing ML deployments [69, 274] and existing guidelines for mitigating AML attacks [120].

An ongoing challenge in AML is the ability to detect when a model is under attack. Knowing this would provide an opportunity to counter the attack before any information is lost or an adverse behaviour is triggered in the model. However, Tramèr [373] has shown that designing techniques to detect adversarial examples is equivalent to robust classification, which is inherently difficult to solve. Adversarial examples may come from the same data distribution on which the model was trained and to which it expects the inputs to belong or may be OUT-OF-DISTRIBUTION (OOD) inputs. Thus, the ability to detect OOD inputs is also an important challenge in AML. Fang et al. [124] established useful theoretical bounds on detectability, particularly an impossibility result when there is an overlap between the in-distribution and OOD data.

Formal methods verification has a long history in other fields that require high assurance, such as avionics and cryptography. Although the results of applying this methodology offer security and safety assurances, they come at a very high cost, which has prevented formal methods from being widely adopted. Currently, formal methods in these fields are primarily used in applications that are mandated by regulations. Applying formal methods to neural networks has the potential to provide much-needed security guarantees, especially in high-risk applications. However, the viability of this technology will be determined by a combination of technical and business criteria — namely, the ability to handle today's complex ML models of interest at acceptable costs. More research is needed to extend this technology to the algebraic operations used in ML algorithms, scale it up to the large models used today, and accommodate rapid changes in the code of AI systems while limiting the costs of applying formal verification.

4.1.3. Evaluation

Another general problem of AML mitigations for both evasion and poisoning attacks is the lack of reliable benchmarks, which causes results from AML papers to be routinely incomparable, as they do not rely on the same assumptions and methods. While there have been some promising developments in this direction [97, 327], more research and encouragement are needed to foster the creation of standardized benchmarks to gain reliable insights into the actual performance of proposed mitigations.

More broadly, the effectiveness of a mitigation is determined not just by how well it will defend against existing attack, but also how well it defends against unforeseen attacks. This means that new mitigations should be tested adversarially, with the researchers proposing the mitigation also trying to break it. This is often difficult and time-consuming, leading to less rigorous and reliable evaluations of novel mitigations; often they appear very powerful, but are quickly shown lack robustness to unforeseen types of attacks.

Finally, this difficulty combines with the difficulty of trading off between different attributes discussed above. Instead of evaluating each attribute in isolation, they should be evaluated simultaneously for any new mitigations, and mitigations should be compared on a Pareto plot (as in Fig. 8) capturing the various tradeoffs that have to be made. This additionally increases the cost to evaluating new mitigations, and can make comparing mitigations difficult - if the green dot represents a new method, it is not possible to say it is an improvement on the red dot, as it is better on one axis but worse on the other.

4.2. Discussion

4.2.1. The Scale Challenge

Data is fundamentally important for training models. Recent trends in GenAI have been towards significant investment in larger models and larger datasets for training them. Few developers of foundation models publish key details about the data sources used in their training [44]. Those who do [247, 371] show the scale of the footprint and the massive amount of data consumed during training. The most recent multi-modal GenAI systems further exacerbate the demand by requiring large amounts of data for each modality.

In most cases, no single entity controls all of the data used to train a particular foundation model. Data repositories are not monolithic data containers but a list of labels and data links to other servers that actually contain the corresponding data samples. This paradigm challenges the classic definition of the corporate cybersecurity perimeter and creates new risks that are difficult to mitigate [57]. Recently published open-source data poisoning tools [241] increase the risk of large-scale attacks on image training data. Although created to enable artists to protect the copyright of their work, these tools may become harmful in the hands of people with malicious intent.

There are several ways this new class of attacks could be mitigated, although it is unclear

how effective mitigations will prove to be as the sophistication of attacks increase. Data and model sanitization techniques (see Sec. 2.3) reduce the impacts of a range of poisoning attacks. They can be combined with cryptographic techniques for origin and integrity attestation to provide assurances downstream, as recommended in the final report of the National Security Commission on AI [267]. Robust training techniques (see Sec. 2.3) offer different approaches to developing theoretically certified defenses against data poisoning attacks with the intention of providing much-needed information-theoretic guarantees for security. The results are encouraging, but more research is needed to extend this methodology to more general assumptions about data distributions, the ability to handle out-of-distribution inputs, more complex models, multiple data modalities, and better performance. Another challenge is applying these techniques to very large models like LLMs and generative diffusion models, which are becoming targets of attacks [55, 90].

4.2.2. Supply Chain Challenges

The literature on AML shows a trend of designing new attacks that are more difficult to detect. Since the poisoning of AI models can persist through safety training and be triggered by attackers on demand [170], significant concerns arise regarding the potential for models to be created with intentional exploits that are hard for organizations deploying and using models to detect. The potential for attacks against open-source dependencies may be particularly acute in the AI context because organizations and researchers may not be able to audit and identify vulnerabilities encoded into a model's weights in the same way it is often possible to audit open-source software. As users come to rely more on the outputs of AI systems — for example, some research suggests that software engineers who over-rely on AI coding assistants' suggestions may produce less secure code [290, 294, 308] — the potential for malicious actors to subtly manipulate the outputs of AI systems may create increased risk.

Additionally, Goldwasser et al. [142] introduced a new class of attacks: information-theoretically undetectable Trojans that can be planted in ML models. If proven practical, the undetectable nature of such attacks would pose significant challenges for AI supply-chain risk management and increase the importance of preventing insider threat throughout the supply chain. DARPA and NIST have also jointly created TrojAI to research the defense of AI systems from intentional, malicious Trojans by developing the technology to detect and investigate these attacks.

4.2.3. Multimodal Models

MULTIMODAL MODELS have shown great potential for achieving high performance on many ML tasks [27, 30, 258, 304, 435]. However, emerging evidence from practice shows that a redundancy of information across the different modalities does not necessarily make the model more robust against adversarial perturbations of a single modality. Combining modalities and training the model on clean data alone does not seem to improve adver-

serial robustness. In addition, adversarial training, which is widely used in single modality applications, may become prohibitively expensive as the number of modality combinations increases. Additional effort is required to benefit from the redundant information in order to improve robustness against single modality attacks [417]. Without such an effort, single modality attacks can be effective and compromise multimodal models across a wide range of multimodal tasks despite the information contained in the remaining unperturbed modalities [417, 424]. Moreover, researchers have devised efficient mechanisms for constructing simultaneous attacks on multiple modalities, which suggests that multimodal models might not be more robust against adversarial attacks despite improved performance [77, 333, 415].

The existence of simultaneous attacks on multimodal models suggests that mitigation techniques that only rely on single modality perturbations are not likely to be robust.

4.2.4. Quantized Models

Quantization is a technique for efficiently deploying models to edge platforms, such as smart phones and IoT devices [138]. It reduces the computational and memory costs of running inference on a given platform by representing the model weights and activations with low-precision data types. For example, quantized models typically use 8-bit integers (int8) or even more compact 4-bit representations instead of the usual 32-bit floating point (float32) numbers for the original non-quantized model.

This technique has been widely used with PredAI and increasingly with GenAI models [108]. However, quantized models inherit the vulnerabilities of the original models and introduce additional weaknesses that make them vulnerable to adversarial attacks. Error amplification from reduced computational precision adversely affects the adversarial robustness of the quantized models. While there are some useful mitigation techniques for PredAI models [216], the effects of quantization on GenAI models have not been studied as thoroughly. Organizations that deploy such models should continuously monitor their behavior. Recent results [118] reveal that widely used quantization methods can be exploited to produce a harmful quantized LLM, even though the full-precision counterpart appears benign, potentially tricking users into deploying the malicious quantized model.

4.2.5. Risk Management in Light of AML

A key question that this taxonomy deliberately leaves aside is how organizations can make decisions about the development and use of AI systems in light of evidence about the increasing diversity of AML attacks and the efficacy and limitations of available mitigations.

Especially in GenAI, some model developers and application builders have moved towards paradigms for testing adversarial risks as part of pre-deployment testing and evaluation

of models, such as through a structured process for RED TEAMING [68, 133]. NIST has produced an initial public draft of guidance for model developers on managing risks associated with the misuse of foundation model capabilities, including through pre-deployment evaluations. [275] NIST [273] and Barrett et al. [28] have also developed risk profiles for generative AI systems that map to the NIST AI RMF [274] that may assist model developers and users in assessing risks, including those from adversarial attacks.

However, a persistent challenge remains in the fact that many AML mitigations are empirical in nature and lack theoretical or provable guarantees. In fact, several research results have pointed to theoretical *limits* on AML mitigations, including the impossibility of model-based detection to prevent all impermissible outputs [140] and findings that, so long as a model has any probability of exhibiting an undesired behavior, there exist prompts that can trigger that behavior, implying that any alignment process that attenuates but does not remove an unwanted behavior will remain vulnerable to adversarial prompting attacks. [406]

These theoretical limitations do not obviate the utility of pre-deployment adversarial testing, since such testing can potentially foreclose many attack vectors and thus increase the difficulty of mounting a successful attack above would-be attackers' threshold of effort or capability. However, they suggest that organizations seeking to manage risks related to the development of models with potentially harmful capabilities or the delegation of trust to models in high stakes contexts may need to consider practices and measures beyond adversarial testing to manage the risks associated with AML attacks.

4.2.6. AML and Other AI System Characteristics

A final consideration with respect to adversarial machine learning, and one closely related to questions of risk management, is how to relate and integrate consideration of AML attacks to definitions and processes relating to other desired AI system characteristics.

For example, managing the security of AI systems will require combining mitigations from the field of AML with best practices for the development of secure software from the field of cybersecurity. Understanding and relating these practices to each other, as well as identifying whether there are other key considerations for AI security that fall outside of the scope of either AML or cybersecurity, will be critical as organizations seek to extend existing cybersecurity processes and best practices to address the security of newly adopted AI systems.

Similarly, robustness to AML attacks may play an important role in areas beyond the remit of security, such as in AI safety [275] or in achieving other characteristics of trustworthy AI systems [274]. AML is neither a complete solution to, nor a subset of, any one of these characteristics, and as such, more precisely relating AML attacks and mitigations to processes for achieving these goals and managing risks in AI systems is an area for ongoing work.

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Appendix A. Glossary

Clicking on the page number at the end of a definition will navigate to the page where the term is used.

A

adversarial example A modified testing sample that induces misclassification or misbehavior of a machine learning model at deployment time. ix, 6

adversarial machine learning Attacks that exploit the statistical, data-based nature of machine learning systems. xii, 1

agent Software programs that can interact with their environment, receive information, and undertake self-directed actions in service of a larger, externally-specified goal. 1, 35, 37, 39, 50, 52, 54

area under the curve A measure of the ability of a classifier to distinguish between classes in machine learning. A higher AUC means that a model performs better when distinguishing between the two classes. AUC measures the entire two-dimensional area under the RECEIVER OPERATING CHARACTERISTIC (ROC) curve. 30

attribute inference attacks An attack against machine learning models that infers sensitive attributes of a training data record, given partial knowledge about the record. 7

availability breakdown In the AML context, a disruption of the ability of other users or processes to obtain timely and reliable access to an AI system's outputs or functionality. 6, 39

B

backdoor pattern A transformation or insertion applied to a data sample that triggers an adversary-specified behaviour in a model that has been subject to a backdoor poisoning attack. For example, in computer vision, an adversary could poison a model such that the insertion of a square of white pixels induces a desired target label. 6, 22, 107

backdoor poisoning attack A poisoning attack that causes a model to perform an adversary-selected behaviour in response to inputs that follow a particular BACKDOOR PATTERN. 6, 42

C

classification The task of predicting which of a set of discrete categories an input belongs to. 5

convolutional neural networks A class of feed-forward neural networks that include at least one convolutional layer, referred to as CNNs. In convolutional layers, feature detectors (known as kernels or filters) detect specific features across the input data. CNNs are primarily used for processing grid-like data, such as images, and are particularly effective for tasks like image classification, object detection, and image segmentation. 5, 31

D

- data confidentiality** A well-established concept in cybersecurity referring to the protection of sensitive information from unauthorized access and disclosure. 7
- data poisoning** A POISONING ATTACKS in which an adversary controls part of the training data. 5, 36, 37, 40, 111
- data privacy attacks** Attacks against machine learning models that extract sensitive information about training data. 7
- data reconstruction** Privacy attacks that reconstruct sensitive data in a model's training data from aggregate information. 7, 28
- deployment stage** The stage of the machine learning pipeline in which a model is deployed into a live or real-world environment for use, such as being integrated into an enterprise application or made available to end users through an API. 5, 37, 38
- diffusion models** A class of latent variable generative models consisting of three major components: a forward process, a reverse process, and a sampling procedure. The goal of the diffusion model is to learn a diffusion process that generates the probability distribution of a given dataset. It is widely used in computer vision on a variety of tasks, including image denoising, inpainting, super-resolution, and image generation. 34
- direct prompt injection** A DIRECT PROMPTING ATTACK in which the attacker exploits PROMPT INJECTION. 43, 110
- direct prompting attack** In the generative AI context, an attack conducted by the primary user of the system through QUERY ACCESS (e.g., as opposed to through RESOURCE CONTROL). 34, 43, 108, 110
- discriminative** A type of machine learning method that learns to discriminate between classes. 5

E

- energy-latency attack** An attack that exploits the performance dependency on hardware and model optimizations to negate the effects of hardware optimizations, increase computational latency, increase hardware temperature, and massively increase the amount of energy consumed. 6, 8
- ensemble learning** A type of a meta machine learning approach that combines the predictions of several models to improve performance. 5
- expectation over transformation** A method for strengthening adversarial examples to remain adversarial under image transformations that occur in the real world, such as angle and viewpoint changes. EOT models these perturbations within the optimization procedure. Rather than optimizing the log-likelihood of a single example, EOT uses a chosen distribution of transformation functions that take an input controlled by the adversary to the "true" input perceived by the classifier. 16

F

federated learning A type of machine learning in which a model is trained in a decentralized fashion using multiple data sources without pooling or combining the data in any centralized location. Federated learning allows entities or devices to collaboratively train a global model by exchanging model updates without directly sharing the data that each entity controls. 5, 31

feedforward neural networks Artificial neural networks in which the connections between nodes is from one layer to the next and do not form a cycle. 31

fine-tuning The process of adapting a pre-trained model to perform specific tasks or specialize in a particular domain. This phase follows the initial pre-training phase and involves further training the model on task-specific data. This is often a supervised learning task. 37

fine-tuning circumvention Fine-tuning to remove model refusal behaviour or other model-level safety interventions. 41

formal methods A mathematically rigorous technique for the specification, development, and verification of software systems. 18

foundation model In generative AI, models trained on broad data using SELF-SUPERVISED LEARNING that can be adapted such as through fine-tuning for a variety of downstream tasks [311]. 111

functional attack An adversarial attack that is optimized for a set of data in a domain rather than per data point. 13, 23

G

generative adversarial networks A machine learning framework in which two neural networks contest with each other in the form of a zero-sum game, where one agent's gain is another agent's loss. A GAN learns to generate new data with the same statistics as the training set. See [143] for further details. 31, 34

generative pre-trained transformer (GPT) A family of machine learning models based on the transformer architecture [383] that are pre-trained through SELF-SUPERVISED LEARNING on large data sets of unlabelled text. This is the current predominant architecture for large language models. 34

graph neural network A neural network designed to process graph-structured data. GNNs perform optimizable transformations on graph attributes (e.g., nodes, edges, global context) while preserving graph symmetries such as permutation invariance. GNNs utilize a “graph-in, graph-out” architecture that takes an input graph with information and progressively transforms it into an output graph with the same connectivity as that of the input graph. 31

H

hidden Markov model A Markov model in which the system being modeled is assumed to be a Markov process with unobservable states. The model provides an observable process whose outcomes are influenced by the outcomes of a Markov model in a known way. An HMM can be used to describe the evolution of observable

events that depend on internal factors that are not directly observable. In machine learning, it is assumed that the internal state of a model is hidden but not its hyperparameters. 31

I

indirect prompt injection A type of PROMPT INJECTION executed through RESOURCE CONTROL rather than through user-provided input as in a DIRECT PROMPT INJECTION. 39–41, 50

integrity violation In the AML context, an AI system being forced to misperform against its intended objectives, producing outputs or predictions that align with the attacker’s objective. 6, 40

J

jailbreak A DIRECT PROMPTING ATTACK intended to circumvent restrictions placed on model outputs, such as circumventing refusal behaviour to enable misuse. 34, 38, 42, 43, 52

L

label flipping A type of data poisoning attack in which an adversary is restricted to changing the training labels. 20

label limit A capability with which an attacker does not control the labels of training samples in supervised learning. 8

logistic regression A type of linear classifier that predicts the probability of an observation being part of a class. 5

M

machine unlearning A technique that involves selectively removing the influences of specific training data points from a trained machine learning model, such as to remove unwanted capabilities or knowledge in a foundation model, or to enable a user to request the removal of their records from a model. Efficient approximate unlearning techniques may not require retraining the ML model from scratch. 33

membership-inference attack A data privacy attack to determine whether a data sample was part of the training set of a machine learning model. 7, 28

misuse enablement In the AML context, a circumvention of technical restrictions imposed by the AI system’s owner on its use, such as restrictions designed to prevent a GenAI system from producing outputs that could cause harm to others. 40

model control A capability with which an attacker can control the machine learning model parameters. 8, 37, 41, 111

model extraction A type of privacy attack that extracts details of the model architecture and/or parameters. 7, 28, 31, 40, 41, 47

model poisoning A POISONING ATTACKS which operates through MODEL CONTROL. 5, 6, 37, 41, 111

model privacy attacks An attack against machine learning models to extract sensitive information about the model. 7

multimodal models A model that processes and relates information from multiple sensory modalities that each represent primary human channels of communication and sensation, such as vision and touch. 58

O

out-of-distribution Data that was collected at a different time and possibly under different conditions or in a different environment than the data collected to train the model. 56

P

poisoning attacks Adversarial attacks in which an adversary interferes with a model during its TRAINING STAGE, such as by inserting malicious training data (DATA POISONING) or modifying the training process itself (MODEL POISONING). 5, 108, 111

pre-training A component of the TRAINING STAGE in which a model learns general patterns, features, and relationships from vast amounts of unlabeled data, such as through SELF-SUPERVISED LEARNING. Pre-training can equip models with knowledge of general features or patterns which may be useful in downstream tasks (see FOUNDATION MODEL), and can be followed with additional training or fine-tuning that specializes the model for a specific downstream task. 37

privacy compromise In the AML context, the unauthorized access of restricted or proprietary information that is part of an AI system, including information about a model's training data, weights or architecture; or sensitive information that the model accesses such as the knowledge base of a GenAI RETRIEVAL-AUGMENTED GENERATION (RAG) application. 7, 40

prompt extraction An attack that tries to divulge the system prompt or other information in the context of a large language model that would normally be hidden from a user. 38, 41

prompt injection An attack which exploits the concatenation of untrusted input with a prompt constructed by a higher-trust party such as the application designer. 38, 41, 108, 110

property inference A data privacy attack that infers a global property about the training data of a machine learning model. 7

Q

query access A capability with which an attacker can issue queries to a trained machine learning model and obtain predictions or generations. 8, 40, 108

R

receiver operating characteristic (ROC) A curve that plots the true positive rate versus the false positive rate for a classifier. 107

red teaming in the AI context, means a structured testing effort, often adopting adversarial methods, to find flaws and vulnerabilities in an AI system, including unforeseen or undesirable system behaviors or potential risks associated with the misuse of the system. [366]. 60

regression A type of supervised machine learning model that is trained on data, including numerical labels (i.e., response variables). Types of regression algorithms include linear regression, polynomial regression, and various non-linear regression methods. 5

reinforcement learning A type of machine learning in which a model learns to optimize its behavior according to a reward function by interacting with and receiving feedback from an environment. 5

resource control A capability in which an attacker controls one or more external resources consumed by a machine learning model at inference time, particularly for GenAI systems such as retrieval-augmented generation applications. 41, 50, 108, 110

retrieval-augmented generation (RAG) A type of GenAI system in which a model is paired with a separate information retrieval system (or "knowledge base"). Based on a user query, the RAG system identifies relevant information within the knowledge base and provides it to the GenAI model in context for the model to use in formulating its response. RAG systems allow the internal knowledge of a GenAI model to be modified without the need for retraining. 1, 35, 37, 38, 40, 46, 50, 111

rowhammer attack A software-based fault-injection attack that exploits dynamic random-access memory disturbance errors via user-space applications and allows the attacker to infer information about certain victim secrets stored in memory cells. Mounting this attack requires the attacker to control a user-space unprivileged process that runs on the same machine as the victim's machine learning model. 31

S

self-supervised learning A type of machine learning that relies on generating implicit labels from unstructured data rather than relying on explicit, human-created labels. Self-supervised learning tasks are constructed to allow the true labels to be automatically inferred from the training data (enabling the use of large-scale training data) and to require models to capture essential features or relationships within the data to solve them. For example, a common self-supervised learning task is providing a model with partial data with the task to accurately generate the remainder. 109, 111

semi-supervised learning A type of machine learning in which a small number of training samples are labeled, while the majority are unlabeled. 5

shadow model A model that imitates the behavior of the target model. The training datasets and the truth about membership in these datasets are known for these models.

Typically, the attack model is trained on the labeled inputs and outputs of the shadow model. 25

side channel Allows an attacker to infer information about a secret by observing the non-functional characteristics of a program (e.g., execution time or memory) or measuring or exploiting the indirect coincidental effects of the system or its hardware (e.g., power consumption variation, electromagnetic emanations) while the program is executing. Most commonly, such attacks aim to exfiltrate sensitive information, including cryptographic keys. 31

source code control A capability with which an attacker controls the source code of a machine learning algorithm. 8

supervised learning A type of machine learning in which a model learns to predict explicit (often human-generated) labels or output values for data. 5

support vector machines Models that implement a decision function in the form of a hyperplane that serves to separate (i.e., classify) observations that belong to one class from another based on patterns of information about those observations (i.e., features). 5, 6, 31

system prompt Application-specific instructions provided in-context to a GenAI system by the model developer or application designer. System prompts are typically prepended to other input, and may be higher-trust than other forms of input. 38, 43, 46, 52

T

targeted poisoning attack A poisoning attack that changes the prediction on a small number of targeted samples. 6, 42

testing data control A capability with which an attacker controls the testing data input to the machine learning model. 8

training data control A capability in which an attacker controls some or all of the training data of a machine learning model. 7, 40

training data extraction The ability of an attacker to extract the training data of a generative model by prompting the model with specific inputs. 7, 46

training stage The stage of a machine learning pipeline in which a model learns parameters that minimize its error against an objective function based on training data. 5, 37, 111

trojan In the machine learning context, a malicious modification to a model that is difficult to detect, may appear harmless, but that can alter the intended function of the system upon a signal from an attacker to cause a malicious behavior desired by the attacker. For Trojan attacks to be effective, the trigger must be rare in the normal operating environment so that it does not affect the normal effectiveness of the AI and raise the suspicions of users. In the machine learning context, trojan may be used interchangeably with backdoor pattern. 2

U

unsupervised learning A type of machine learning in which a model learns based on patterns in unlabeled data, such as learning a function to cluster or group data points.

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